4 THE SHEAR STRENGTH OF ROCKFILL

4.1 OUTLINE OF THIS CHAPTER

The work by Marsal (1973) on the shear strength of rockfill showed that the strength of rockfill may vary directly with normal effective stress, dry density, particle roughness, particle crushing strength and inversely with grain size, uniformity of grading, and particle shape.

The aim of this Chapter is to verify and extend this work using an extensive literature review and an analysis of a triaxial shear strength database which has been collected from the literature, and from organisations who have carried out testing for dams and other projects.

In the context of this Chapter the term rockfill encompasses both material of alluvial origin and that of blasted quarry origin. The differentiation between the two can be made by looking at parameters such as angularity.

This Chapter also comprises part of the larger study that is investigating rock mass strength. The author believes that it is reasonable to assume that a compacted rockfill is representative of a poor quality rock mass as defined by Hoek and Brown (1980). The author has therefore used the data collected in this report to create a lower bound for the Hoek-Brown criterion.

This Chapter presents a summary of the main factors affecting rockfill strength from the literature; the triaxial shear strength database; statistical results from the database; and several shear-strength criteria for rockfill using Hoek-Brown and other methods.

4.2 FACTORS AFFECTING THE SHEAR STRENGTH OF ROCKFILL

The author believes that the shear strength of rockfill at a particular confining stress may be seen as the combination of a basic friction angle (shearing between rock surfaces), ϕ_b, plus a dilation component less an amount caused by asperity or particle crushing/shearing and reorientation of particles. In terms of friction angle:

$$f = f_b + i - f_{\text{crush}} \tag{4.1}$$

Dilation ceases to occur at the critical confining pressure as defined by Seed and Lee (1967).

Marsal (1973) found that the shear strength of rockfill may vary directly with normal effective stress, dry density, particle roughness, particle crushing strength and inversely with grain size, uniformity of grading, and particle shape. The following discussion centres on what factors have been found to affect the shear strength of rockfill. An attempt is made to relate these factors to the equation above.

4.2.1 Confining Pressure

Several authors have shown the shear strength curve for rockfill is non-linear, particularly at low confining pressures (triaxial: Leslie, 1963, Marachi et al, 1969, Leps, 1970, Bertacchi & Bellotti, 1970, Penman et al., 1982 and Indraratna et al., 1993; direct shear: Dobr & Rozsypal, 1974 and Anagnosti & Popovic, 1982; plane strain: Al-Hussaini, 1983). This would be consistent with the theory that as normal stress is increased dilation is suppressed and therefore shear strength increase is reduced. Indeed, Boughton (1970) found "some indication … that the limiting value of ϕ as σ'_3 increases is the value for the surfaces of the individual rock pieces" or ϕ_b in Equation 4.1.

The effect of a curved strength envelope has a large impact on the stability analysis of rockfill dams where shallow slip surfaces are shown by the analysis to be critical using traditional average *c*-ϕ analysis. These failure surfaces are not found in practice and this

is likely to be due to the high frictional strength of rockfill at the low confining pressures acting near the surface.

Figure 4.1 shows a curved envelope together with various *c*-ϕ representations. The tangent friction angle, ϕ_t, is defined as the slope of the tangent to the curved shear strength envelope at a given normal stress. The secant friction angle, ϕ_{sec}, is defined as the slope of the tangent from the origin to the Mohr's circle for a particular normal stress (Equation 4.2). It is this secant friction angle that is most commonly quoted when test results are published.

$$f_{sec} = \sin^{-1}\left(\frac{s_1 - s_3}{s_1 + s_3}\right) \tag{4.2}$$

Where the Mohr-Coulomb parameters are taken as $c = 0$ and $\phi = \phi_{sec}$ the shear strength will be underestimated for normal stresses less than s_{n2} and over estimated for stresses greater than s_{n2}. Where the Mohr-Coulomb parameters are taken as $c = c_t$ and $\phi = \phi_t$ the strength will be overestimated at confining pressures less than the normal stress at the tangent point. Where c_t is taken as zero, and $\phi = \phi_t$, as sometimes is done by designers, the strength will be significantly underestimated. This shows the importance of using a non-linear strength criterion or stress dependent parameters e.g. a bi-linear strength envelope.

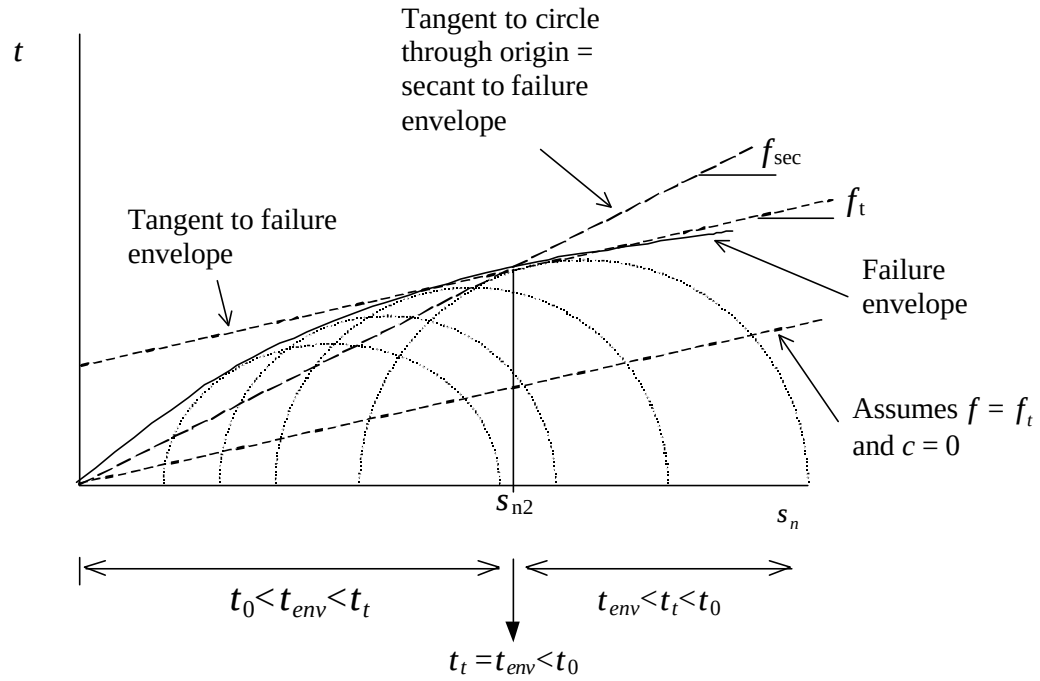


Figure 4.1. Methods for representing the shear strength envelope

As is typical in the literature, this report uses $\phi = \phi_{sec}$ in its discussion unless otherwise stated.

Marachi et al (1969) carried out large scale triaxial tests on highly angular argillite, crushed basalt and rounded amphibolite and found that ϕ does not appear to decrease significantly beyond σ_n = 4.5MPa. Note that they did not perform tests beyond a confining pressure of 4.5MPa. Figure 4.2 and Figure 4.3 show curved strength envelopes from triaxial and direct shear tests respectively which support this.

Indraratna and his co-workers (Indraratna et al., 1993, 1998, Indraratna, 1994) performed tests on greywacke rockfill and basalt ballast. They found that the shear strength of the failure envelope was highly curved for confining stresses of less than 500kPa. Penman et al (1982) found a similar result for confining stresses below 400kPa. Indraratna found that the shear strength could be approximated by a linear Mohr-Coulomb criterion at stresses higher than 1.5MPa (as compared to the Marachi value of 4.5MPa above). Figure 4.4 shows the variation of ϕ_{sec} with σ_n on a log scale presented by Indraratna et al. (1993). The figure also shows a lower limit to the shear strength of rockfill proposed by Indraratna et al (1993).

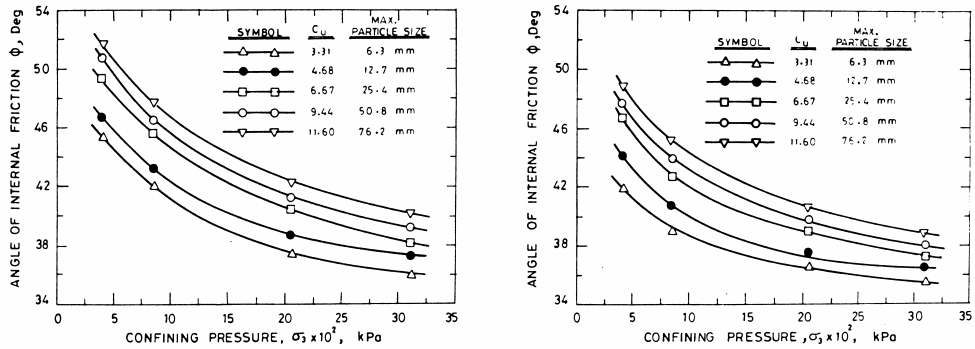


Figure 4.2. Variation of secant friction angle, f_{sec}, with respect to cell confining stress, s_3, for (a) dense and (b) medium dense crushed basalt from triaxial tests (Al-Hussaini, 1983)

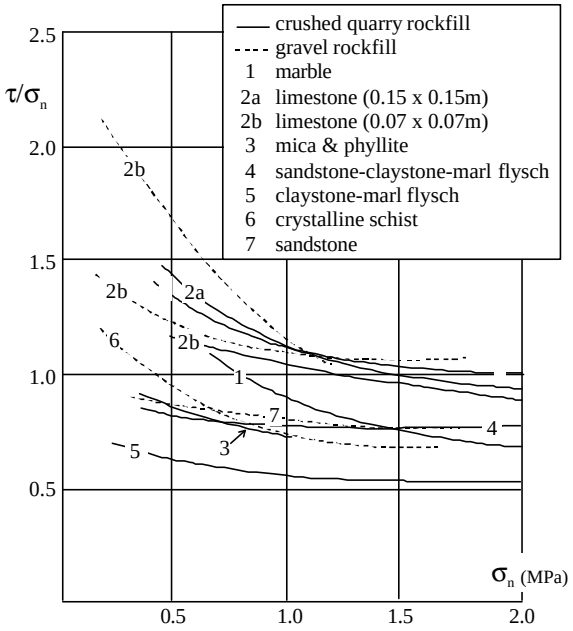


Figure 4.3. Average strength of rockfills from large-scale direct shear tests (Anagnosti & Popovic, 1982)

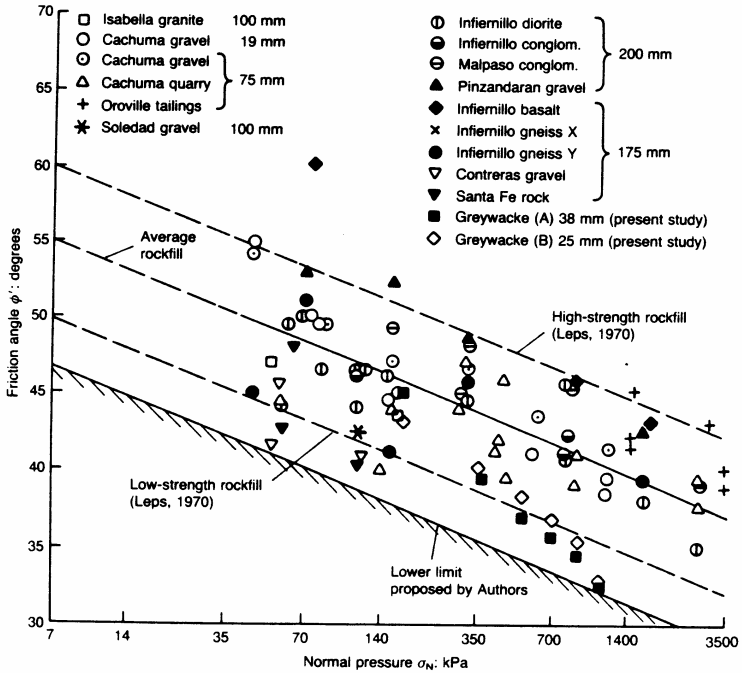


Figure 4.4. Variation of secant friction angle, f_{sec}, with normal stress s_n (Indraratna et al., 1993)

4.2.2 Particle Strength

The strength of the individual particles could be expected to have some effect on the shear strength of rockfill. Where the confining pressure, σ'_3, is low compared to the particle (or unconfined compressive) strength, σ_c, it could be expected that variations in particle strength would have minimal effect on the shear strength. The only exception to this is if it could be shown that σ_c affects ϕ_b. It has been shown however that ϕ_b generally falls within a narrow range for defects in rocks which have not been tectonically sheared. As σ'_3/σ_c decreases, dilation will be the main contributor to increased shear strength. As σ'_3/σ_c increases, failure of particles could be expected through crushing shearing and/or splitting, and thus a reduction in dilation would take place leading to lower shear strength.

Particle strength should therefore play a major role with higher strength particles generally leading to higher strength rockfill. Hirshfeld et al (1973) found the curvature of the shear strength envelope to be more distinct for weak particles. This would be due to the rapid reduction in dilation with increase in confining stress. A reduction in dilation for high strength particles could be expected at much higher stresses than weak particles.

Anagnosti and Popovic (1982) found for crushed rocks, there is a large drop in shear strength up to normal stresses of 1MPa for strong rockfills (Figure 4.5) with a much smaller drop for weaker flysch materials (Figure 4.6). It is likely that tests were not carried out at low enough confining pressures to capture the curvature in the flysch material.

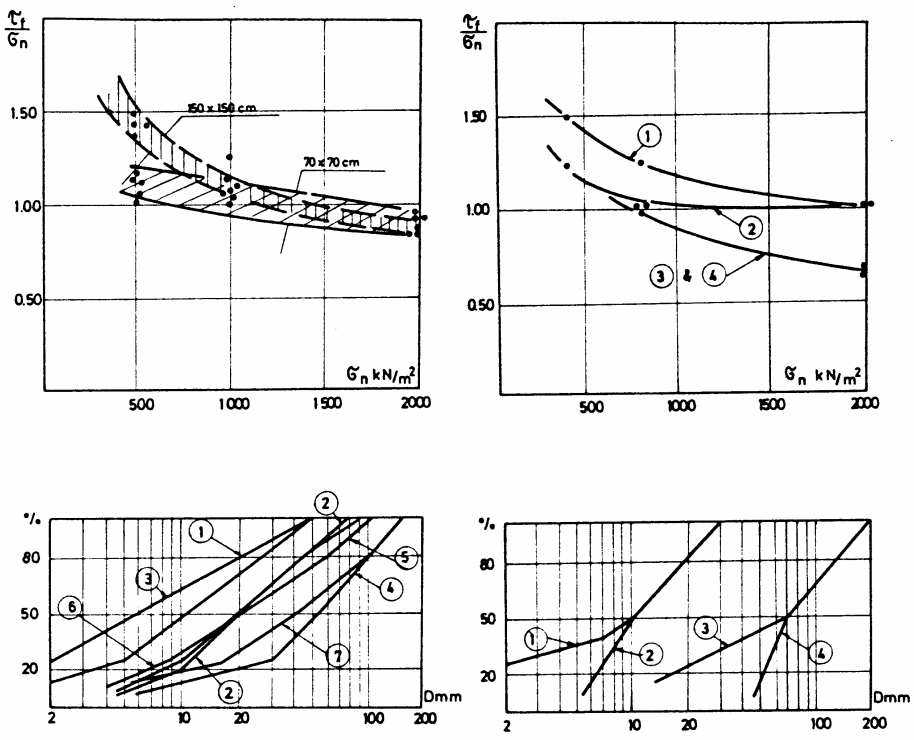


Figure 4.5. Shear strength and grain size curves for crushed (a) limestone and (b) marble (Anagnosti and Popovic, 1982)

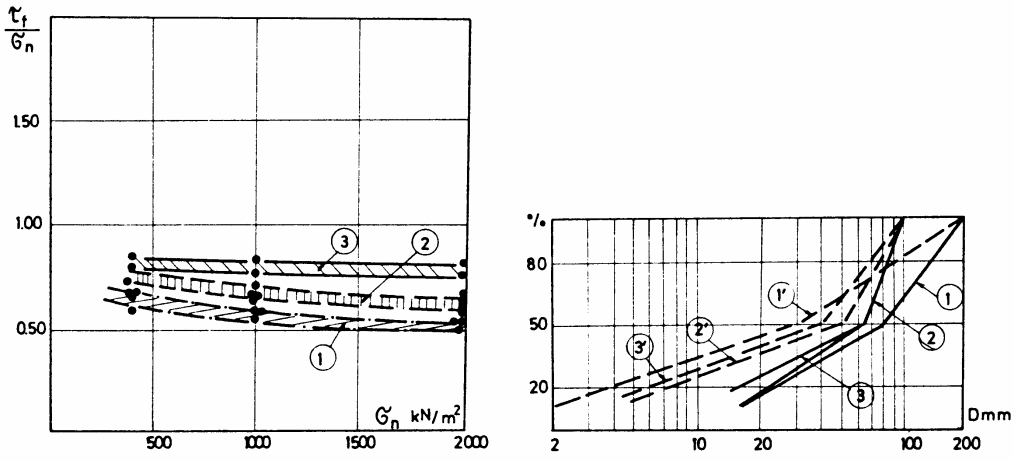


Figure 4.6. Shear strength and grain size curves for crushed flysch sandstone-marl rockfill (Anagnosti and Popovic, 1982)

4.2.3 Uniformity Coefficient

Generally, it could be expected that a poorly-graded rockfill (low uniformity coefficient, c_u, Equation 4.3), would have a higher strength than a well-graded rockfill assuming a constant void ratio for both. A well-graded material would be more likely to reduce the amount of dilation required due to the 'gaps' in the gravel matrix being filled with smaller particles. However, Marachi et al (1969) claim that if both rockfills were compacted to their maximum density then the well graded material could be expected to be stronger as it would have the greater density.

$$c_u = \frac{d_{60}}{d_{10}} \tag{4.3}$$

Where, d_{60} is the particle size for which 60% is finer.

Chiu (1994) using data by Marsal (1967) found that the effect of c_u on the shear strength of rockfill was more pronounced for low σ'_3, with little effect at high σ'_3. c_u affects dilation however, at higher σ'_3 that dilation may have been retarded. Sarac & Popovic (1985) found that the failure envelope was more curved for narrowly graded materials. Al-Hussaini (1983) found strength increased with c_u but felt it may also have been affected by maximum particle size. Anagnosti & Popovic (1982) found for gravels there is a marked drop in strength for the high strength limestone below σ'_3=1MPa possibly due to dilational effects below σ'_3=0.4MPa caused by the uniform grading (Figure 4.7). The well-graded and weaker material does not show this effect (Figure 4.8).

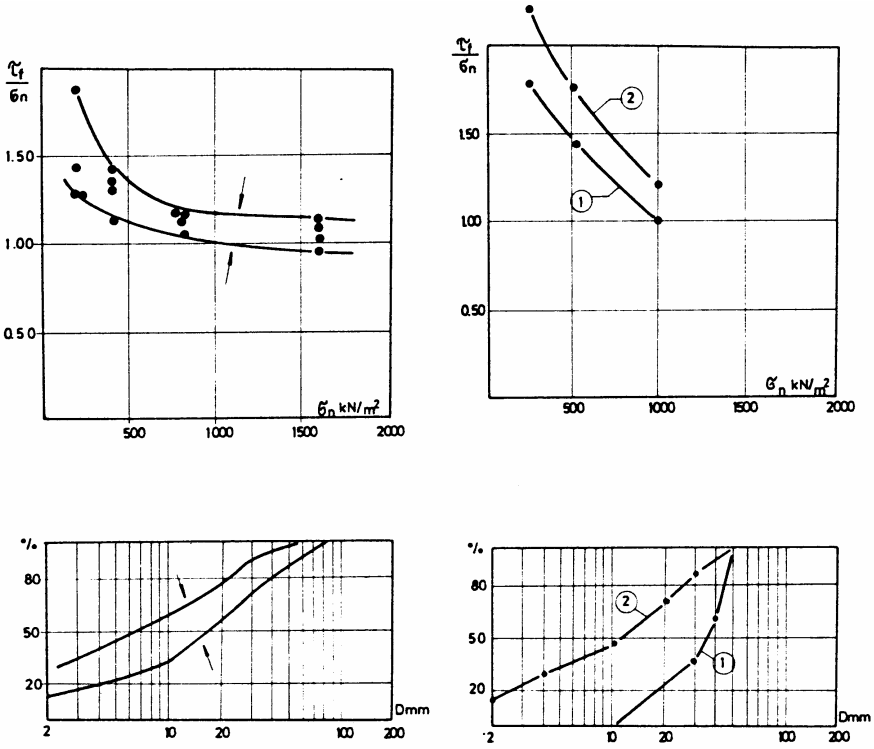


Figure 4.7. Shear strength and grain size curves for different gradings of limestone gravel (Anagnosti and Popovic, 1982)

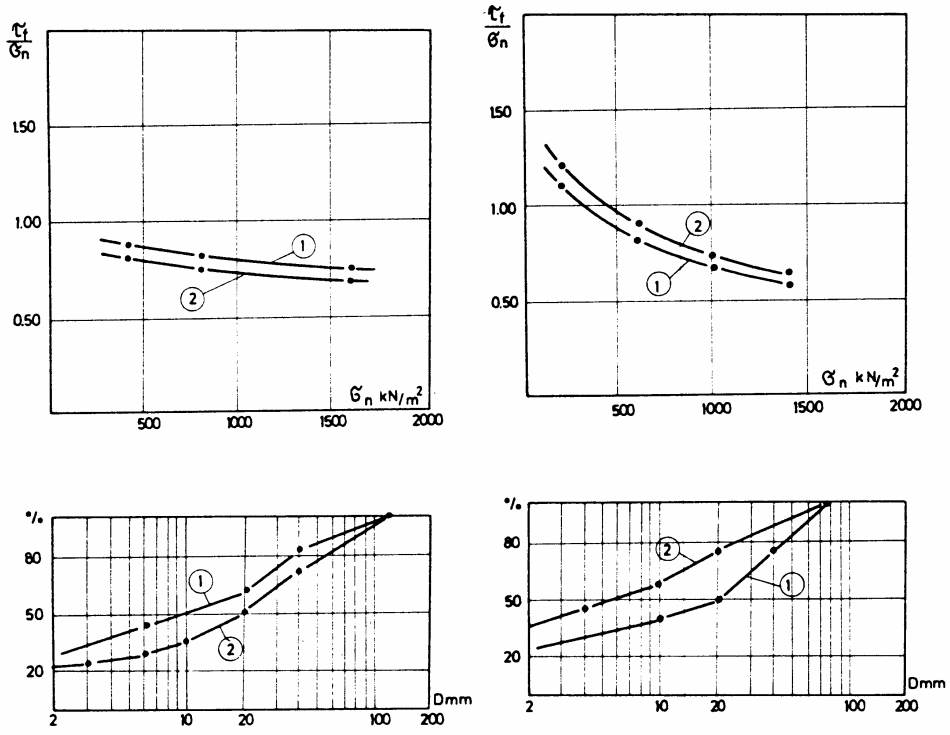


Figure 4.8. Shear strength and grain size curves for (a) crystalline schist and (b) sandstone gravels (Anagnosti and Popovic, 1982)

4.2.4 Density

When comparing the degree of compaction of rockfill relative density or void ratio can be used. Where materials of the same rock type and grading characteristics are tested dry density may be used as it will be related to relative density. Dry density or void ratio will not be a good parameter to use where different types and gradings of rockfills are compared.

Rockfill with a higher relative density has been found to have a higher shear strength (Boughton, 1970, Leps, 1970, Marsal, 1973, Williams & Walker, 1984, Sarac & Popovic, 1985, Tosun et al., 1999). The shape of the Mohr-Coulomb failure envelope is also affected. Initially dense rockfill shows a marked curvature showing a distinct drop in the friction angle whereas an initially loose sample shows minimal curvature and drop in friction. The two curves tend to merge at very high confining pressures. This is consistent with the behaviour of granular soils which reach the critical state or constant volume/void ratio at large strains. Marachi et al (1969) indicate that the curves continue as a straight line whose projection passes through the origin. This is likely due to the dense material requiring dilation at low stresses to fail. At higher confining stresses dilation is restrained and particle shearing/crushing occurs resulting in a lower angle of friction. Initially loose material will require much less dilation as particles have more freedom to move (e.g. rotation) during shearing. Thus without the dilation there will be a minimal drop in the friction angle.

Al-Hussaini (1983) found that dense crushed basalt has a higher strength than medium dense crushed basalt. Dobr & Rozsypal (1974) tested basalt at different dry densities in direct shear. They found that the strength increases with density but, the strength difference reduces with confining pressure. Chui (1994) also found little difference at high confining pressures. Alva-Hurtado et al. (1981) found the opposite with the difference in strengths increasing with confining pressure.

Zeller et al (1957), as reported in Marachi et al (1969), performed triaxial tests on scalped rockfill samples for Goscheoenalp Dam in Switzerland. The confining pressure for all tests was 88kPa. It should be noted that the scalping method used did not result in parallel gradations. Zeller's results (Figure 4.9 and Figure 4.10) show a decrease in

friction angle of approximately 50° to 40° for an initial porosity increase of 8% for all dry rockfill samples (d_{max} of 10mm to 100mm).

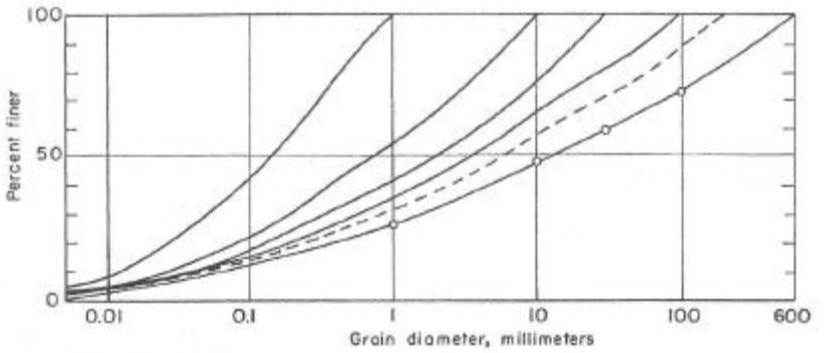


Figure 4.9. Scalped rockfill gradings (Marachi et al, 1969)

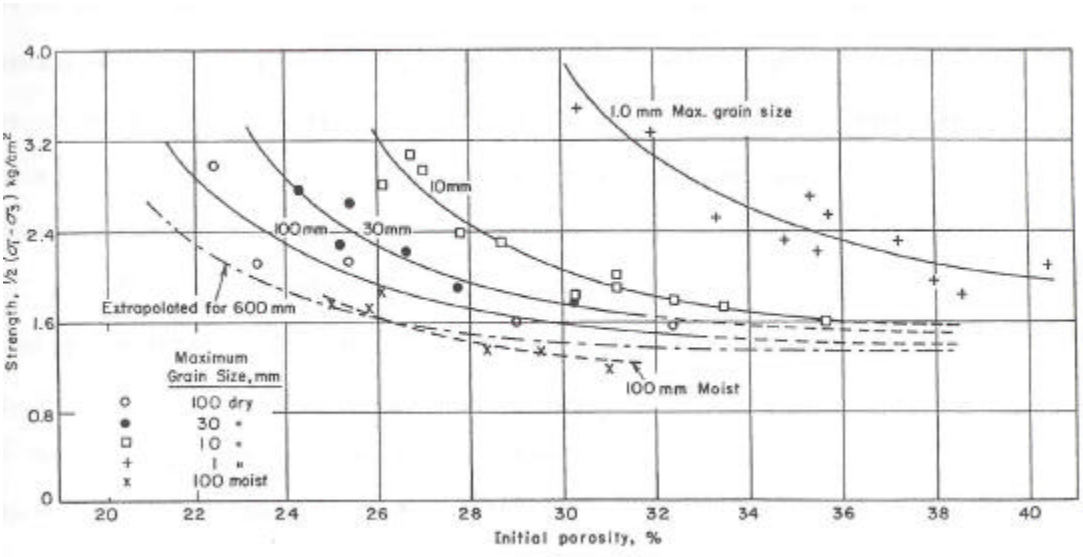


Figure 4.10. Strength porosity relationships with σ_3 = 88kPa (Marachi et al, 1969)

Marachi et al (1969) tested rockfill material from Pyramid Dam. They found a drop in friction angle of approximately 4° for an increase in void ratio of 0.3.

Nakayama et al. (1982) found that the friction angle was reduced with an increase in the void ratio (Figure 4.11). There is no indication in the paper as to what the confining pressure or method of assessing ϕ was. Boughton (1970) and Anagnostic & Popovic (1985) found a similar result although, as might be expected, Anagnostic & Popovic (1985) found the effect was more noticeable at low confining pressures and low C_u.

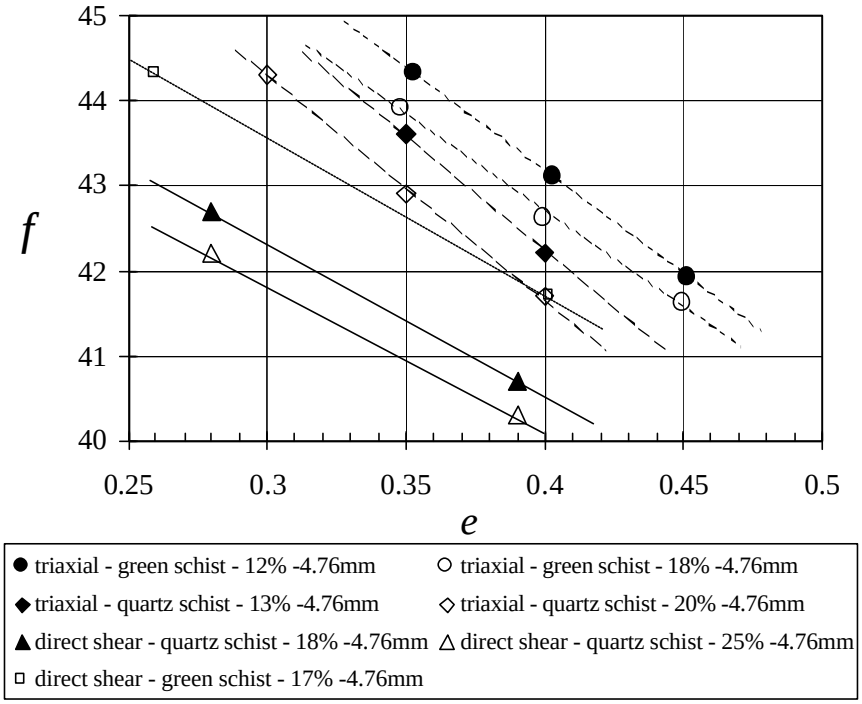


Figure 4.11. Void ratio vs angle of friction (modified from Nakayama et al., 1982)

4.2.5 Maximum Particle Size

There is some conjecture as to the effect of the maximum particle size on the shear strength of rockfill. It is generally accepted that the shear strength decreases with particle size (Marachi et al., 1969 & 1972, Marsal, 1973, Chui, 1994). However, some researches claim no effect (Charles et al., 1980) or the opposite effect (Anagnosti & Popovic, 1981). Some data from Marsal is tabulated below.

The effect of maximum particle size should be considered as two issues:

(I) the effect of increasing maximum particle diameter, d_{max} with constant sample diameter, D; and

(II) the effect of increasing d_{max} with constant d_{max}/D ratio.

(both assuming constant sample height to sample diameter, H/D ratio)

4.2.5.1 Increasing d_{max} with Constant D

Marsal (1973) performed tests on basalt rockfill (the results of the tests are tabulated below). The results show an increase in friction angle of 3-4% for a change in d_{max}/D from 0.07 to 0.18. At high normal stresses (3.9MPa) the effect is more limited (0.3°).

Table 4.1. Increase in f with d_{max}/D from Marsal (1973) data for different s_n

Source	Rock Type	D (mm)		C_u	C_c	σ_n (MPa)	d_{max1}/D	d_{max2}/D	$(\phi_2 - \phi_1)/\phi_2$ (%)
Marsal (1973)	Basalt	1130	A	11 - 18	5.21 – 0.55	0.8	0.07	0.18	3
Marsal (1973)	Basalt	1130	A	11 - 18	5.21 – 0.55	1.6	0.07	0.18	4
Marsal (1973)	Basalt	1130	A	11 - 18	5.21 – 0.55	3.9	0.07	0.18	0.3

Note: d_{max1}, ϕ_1 and d_{max2}, ϕ_2 are the maximum particle sizes and secant friction angle for sample one and sample two respectively.

4.2.5.2 Increasing d_{max} with Constant d_{max}/D

Thiers & Donovan (1981) present Figure 4.12 to estimate the shear strength of rockfill at field scale. The figure plots the angle of internal friction versus maximum particle size. It should be noted that all the samples tested had a constant d_{max}/D ratio.

Table 4.2. Reduction in f with particle size, d_{max}, from Marsal (1973) data

Source	Rock Type	d_{max}/D		C_u	C_c	σ_n (MPa)	d_{max1} (mm)	d_{max2} (mm)	$(f_1 - f_2)/f_2$ (%)
Marsal (1973)	Silicified conglomerate	.18	A	10	1.72	3.92	38.1	200	7
Marsal (1973)	Granitic-gneiss	.18	SA	14	1.30	3.92	38.1	200	20

Note: d_{max1}, ϕ_1 and d_{max2}, ϕ_2 are the maximum particle sizes and secant friction angle for sample one and sample two respectively.

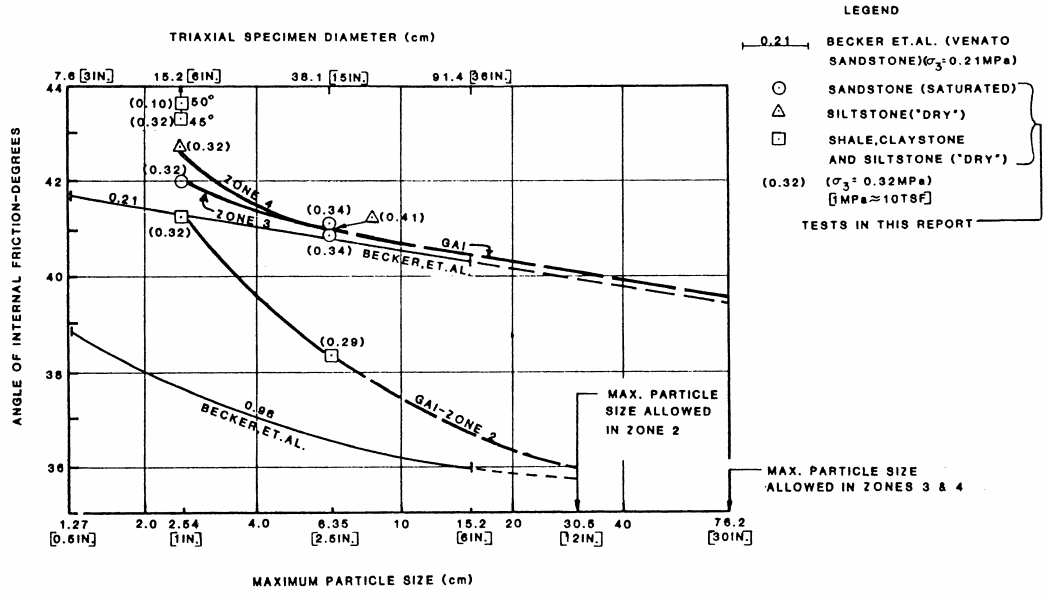


Figure 4.12. Friction angle vs maximum particle size (Thiers & Donovan, 1981)

4.2.6 Silt and Sand Fines versus Gravel and Larger Particle Content

The percent of material passing 2mm is believed to affect the shear strength of rockfill. At the point where the failure shear surface does not incorporate gravel to gravel contact it could be expected that the shear strength of the finer particles would be the sole control on the shear strength of the material. Large triaxial tests for silty gravels were carried out by the USBR (1966). Tests were carried out at 0, 35, 50 and 65% gravel content. Shear strengths were found to be similar for the 0 and 35% gravel content tests. Shear strengths increased considerably at 50% and 65% gravel content. Further tests on clayey gravels found that the shear strength increased considerably between 42 and 50% gravel content. Other conclusions that were reached included:

- Clayey and silty gravel show the shear strength is unaffected by gravel contents below 35%.
- Sandy gravel shows an increase in gravel gives an increase in strength up to a maximum of 50% gravel
- At 50% gravel content, silty gravel and clayey gravel strengths significantly increase.
- At 65% gravel content the silty gravel has similar shear strength to the sandy gravel.

Data from USBR (1966) and USBR (1961) for silty gravel and clayey gravel are plotted on Figure 4.13 and Figure 4.14 respectively. Trend lines (logarithmic) have been plotted on the curves to show the general trends of the data. They should not be used for design.

Marsal (1976) reports two triaxial tests each on rockfill-silt and rockfill-sand mixtures and compares these to a test on rockfill only. The clean rockfill and 10% sand-rockfill mixtures had a ϕ of 34.1° whilst the 30% sand-rockfill had a ϕ of 39°. Marsal (1976) attributes the difference to the lower initial void ratio in the 30% sand-rockfill mixture. The 10% silt-rockfill mixture showed a decrease in ϕ to 28.8° whilst the 30% silt-rockfill mixture had the strength properties of the silt.

Nakayama et al. (1982) performed triaxial and direct shear tests on schist gravels (quartz schist and green schist) with maximum particle size of 63.5mm at Inamura Rockfill Dam. It was found that the friction angle was reduced with an increase in the percent of –4.76mm particles (Figure 4.15).

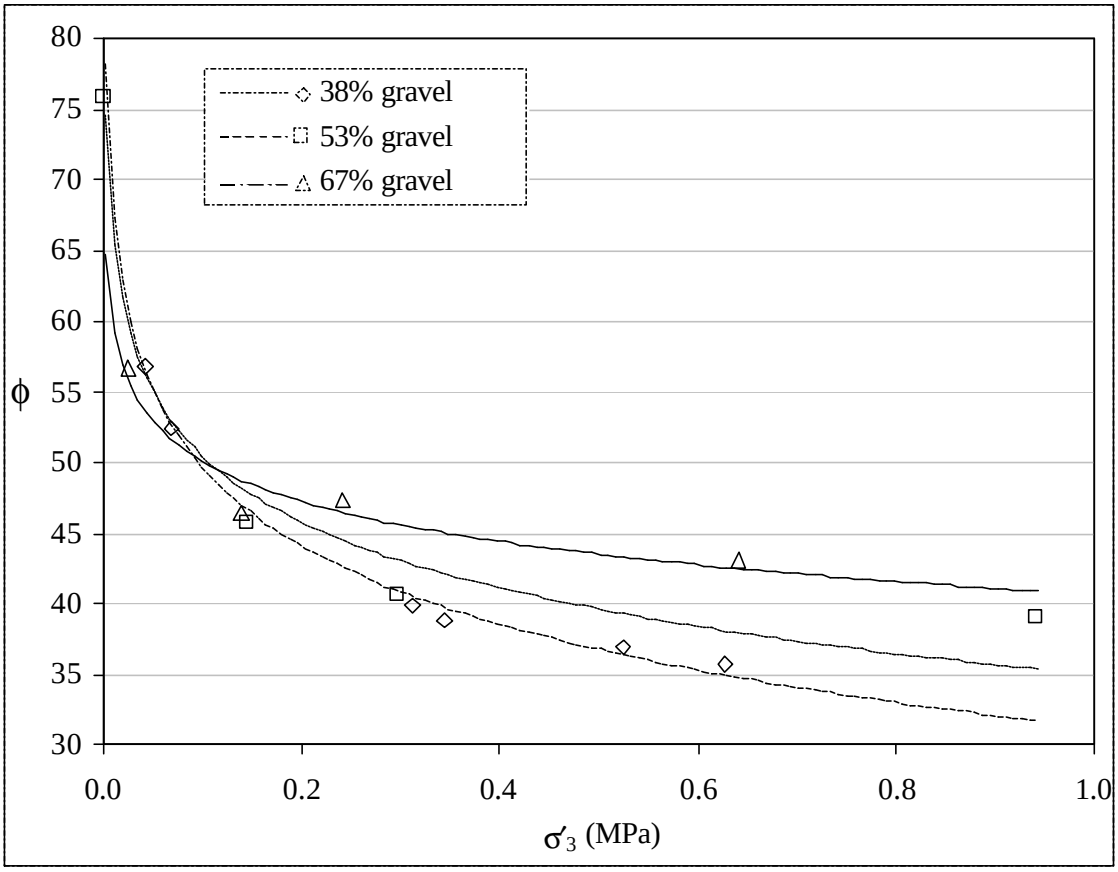


Figure 4.13. Effect of gravel content on f for silty gravel based on USBR (1966)

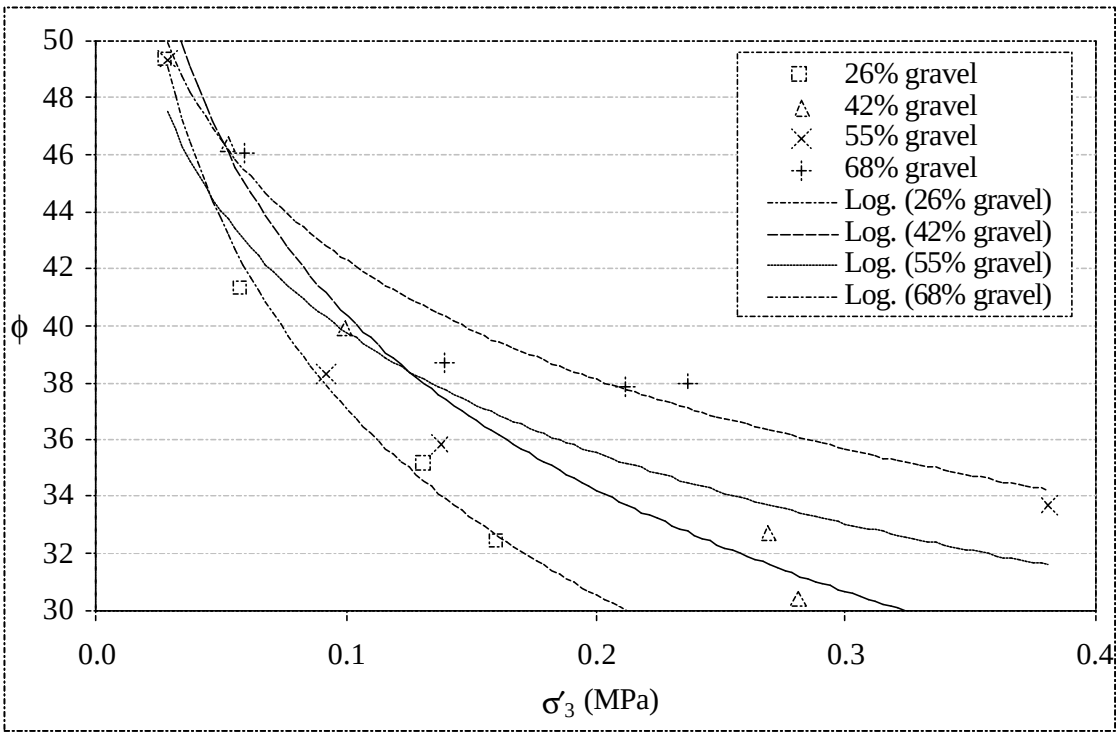


Figure 4.14. Effect of gravel content on f for clayey gravel based on USBR (1961)

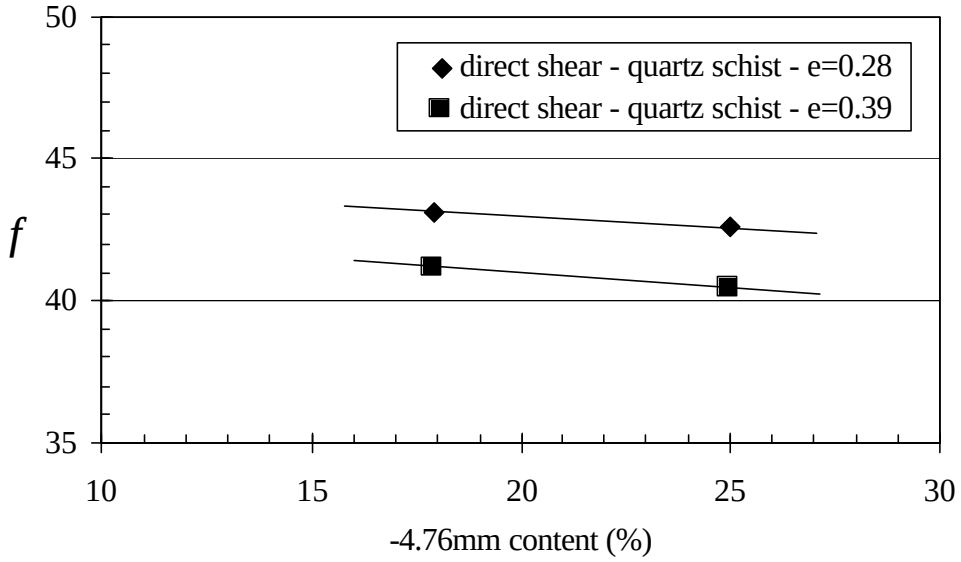


Figure 4.15. –4.76mm content vs secant angle of friction (Nakayama et al., 1982)

4.2.7 Particle Angularity

The strength of highly angular rockfill could be expected to be higher than that for a rounded rockfill at similar relative density and low confining pressures. This is due in some part to the interlocking effect of the angular particles and also increased dilation. At high stresses the effect may not be as prominent and may be the opposite. An angular rockfill may allow for stress concentrations that cause breakage of the particles at high confining pressures reducing dilation and leading to a lower overall rockfill strength than the rounded particles with less stress concentrations.

The shape co-efficient, C_f, is often used to describe the shape of particles and is given in Equation 4.4. The shape coefficient is the volume of n particles of gravel over the equivalent volume of *n* spheres of diameter *D*.

$$C_f = \frac{\sum_0^n V}{\sum_0^n \frac{p}{6} D^3} \tag{4.4}$$

Bertacchi & Berlotti (1970) found that serpentinite (UCS = 100-150MPa, laminar surfaces, sharp edges, C_f = 0.11) gives higher friction angles than tonalite (UCS = 150MPa, irregular rough surfaces, C_f = 0.17) due to the irregularity in the shape of the serpentinite particles. Sarac & Popovic (1985) found a similar result but the effect was less with higher confining pressures.

4.2.8 Other Factors

The type of shear test is believed to have an influence over results. Generally plane strain tests in the literature (Al-Hussaini, 1983 – dense crushed basalt, Barton & Kjærnsli, 1981, Charles & Watts, 1980) give higher friction angles, ϕ'_{ps}, than triaxial shear tests, ϕ'_{ax}. Tests by Marsal et al (1967) and Marsal (1973) discussed by Charles (1991) showed ϕ' at failure is greater in the plane strain tests by up to 8° for the same minor principal stress. The Danish Code of Practice (Steensen-Bach, 1989) gave:

$$f'_{ps} = 1.1 f'_{ax} \tag{4.5}$$

Whilst Wroth (1984) gives:

$$8 f'_{ps} = 9 f'_{ax} \tag{4.6}$$

McWilliam (2001), working with the author, used published data to show that the ratio between ϕ'_{ps} and ϕ'_{ax} varied from 1.02 to 1.1 (c.f. 1.125 from equation 4.6). The ratio varied with material type and confining pressure.

The triaxial test provides a correct value of ϕ as it incorporates the intermediate principal stress, σ'_2 ($=\sigma'_3$). In a plain strain test, strain is prevented in the direction of σ'_2. Where the popular Mohr-Coulomb criterion is used the intermediate stress is ignored which leads to a miscalculation of ϕ. It is a common misconception that using the Mohr-Coulomb derived ϕ from plain strain tests gives a better correlation to field values and should therefore be used in preference to the ϕ from triaxial tests. The backanalysis of failures have traditionally been carried out in two dimensions. This procedure ignores σ'_2 in the same manner as in the derivation of ϕ from the plane strain lab tests. Thus, the plain strain ϕ and the back calculated ϕ are both incorrect but, as they have similar errors in their derivation they tend to give similar results. The author recommends the use of the ϕ derived from triaxial tests for design.

The shear strength of rockfill may decrease with moisture content (Frassoni et al., 1982, Chui, 1994, Marsal, 1967). This is probably related to the drop in unconfined compressive strength for some intact rocks.

4.2.9 Summary of Factors Affecting the Secant Friction Angle

Table 4.3 shows a summary of the factors affecting the secant friction angle discussed in this section. The most significant effects on the secant friction angle appear to be caused by confining pressures, density and maximum particle size. There is also a substantial effect on ϕ_{sec} at high percentages of material finer than gravel. Minor effects on the secant friction angle are caused by angularity and the uniformity coefficient.

Table 4.3. Summary of factors affecting the secant friction angle

Parameter	Effect on ϕ_{sec} with increase in parameter	Comment
Confining pressure	Decrease	Significant effect. The rate of decrease in ϕ_{sec} will drop with increasing confining pressure
Unconfined compressive strength of intact rock	Increase	Effect will depend on the ratio of confining stress to compressive strength
Uniformity coefficient	Decrease	Minor effect and may reverse if samples are compacted to their maximum density prior to testing
Density	Increase	
Maximum particle size (assuming the ratio of maximum particle size to sample diameter is constant)	Decrease	
Ratio of maximum particle size to sample diameter	Increase	
Angularity	Increase	The effect will be most noticeable with highly angular material
Percent finer than gravel size in sample	Decrease	The effect will not be significant at low percentages. At high percentages strength will approach that of the finer material.

4.3 SHEAR STRENGTH CRITERIA

Table 4.5 shows a number of rockfill shear strength equations from the literature. The equations are numbered in the first column of the table. The criteria are empirical having been based on laboratory tests of rockfill.

De Mello (1977) proposed Equation 4.8 as suitable for representing the curved strength envelope. Charles & Watts (1980) also used this form of equation. Indraratna et al. (1993) developed a non-dimensional form of Equation 4.8 for both shear stress and principal stress plots (Equations 4.9 and 4.11). Charles (1991) gives values for the constants A and B in the De Mello (1977) equation (Table 4.4).

Table 4.4. Parameters obtained using De Mello (1977) (Charles, 1991)

	I_D	A	B
Sandy gravel	.95	4.4	.81
Soft rockfill	.95	4.2	.75
Soft rockfill	.70	1.4	.90

Indraratna et al. (1998) also uses Equation 4.12 to describe the shear strength of basalt ballast.

Doruk (1991) created a modified version of the Hoek-Brown criterion by removing the *s* component and thus rendering the unconfined compressive strength of the rock mass to zero. (Equation 4.13)

Sarac & Popovic (1985) analysed a large number of large scale (0.7x0.7x0.4 to 1.9x2.9x1.5m) direct shear tests carried out at the Institute for Geotechnics and Foundation Engineering, Sarajevo (Equation 4.10). The materials tested were generally limestone, sandstone, serpentinite and slate and were from a mixture of sources including: natural; rock debris; and quarry rockfill. The tests were generally carried out for materials for embankment dams over stress ranges between 0.05 to 2.0MPa. Figure 4.17 shows the test results.

Table 4.5. Various shear strength criteria for rockfill

Eq	Reference	Equation	Parameters	Development
4.8	De Mello (1977) Charles & Watts (1980)	$t = As_n^B$	*A, B* see Table 4.4	Empirical curved envelope.
4.9	Indraratna et al (1993) Indraratna (1994)	$\frac{t}{s_c} = a\left(\frac{s_n}{s_c}\right)^b$	a,b = 0.25,0.83 (lower bound, σ_n=0.1-1MPa) a,b = 0.71,0.84 (upper bound, σ_n=0.1-1MPa) a,b = 0.75,0.98 (lower bound, σ_n=1-7MPa) a,b = 1.80,0.99 (upper bound, σ_n=1-7MPa)	Non-dimensionalised form of (4.8). Note that if *A* in (4.8) is independent of σ_c then *a* is not independent of σ_c.
4.10	Sarac & Popovic (1985)	$t_{max} = A(s_n/s_0)^B$	*A* increases with $\sigma_c\uparrow$, $C_u\uparrow$, $\gamma\uparrow$, $d_{50}\uparrow$ ≈ 0.7 to 1.5 (Figure 4.19,Figure 4.20) *B* increases with $\sigma_c\uparrow$, $C_u\uparrow$, $\gamma\downarrow$ ≈ 0.419 to 0.911 (Figure 4.18) σ_0 = 1MPa	Developed from large-scale direct shear tests (up to σ_n = 2MPa) on quarry rockfill and natural gravels. Various sedimentary rocks were used.(**Figure 4.17**)
4.11	Indraratna et al (1993) Indraratna (1994)	$\frac{s_1'}{s_c} = a\left(\frac{s_3'}{s_c}\right)^b$	α,β = 0.4,0.62 (lower bound, 0.1 to 1MPa) α,β = 0.78,0.65 (upper bound, 0.1 to 1MPa) α,β = 2.71,0.96 (lower bound, 1 to 7MPa) α,β = 3.58,0.90 (upper bound, 1 to 7MPa)	Alternative of (4.9) for principal stresses
4.12	Indraratna et al (1998)	$\frac{s_1'}{s_3'} = as_3'^b$	*a, b* = 84.98,-0.49 (gradation A) *a, b* = 125.17,-0.56 (gradation B)	Developed empirically for two gradations of ballast Undefined at $\sigma'_3 = 0$
4.13	Doruk (1991)	$s_1' = s_3' + \left(\frac{ms_3'}{s_c}\right)^a s_c$	*m, a*	Developed from Hoek-Brown by setting the rockfill compressive strength to zero (i.e. *s* = 0).

Eq	Reference	Equation	Parameters	Development
4.14	Barton & Kjærnsli (1981)	$\phi' = R\log\left(\frac{S}{\sigma_n}\right) + \phi_b$	ϕ' - arctan τ/σ_n R - equivalent roughness S - equivalent strength ϕ_b - basic friction angle – from tilt tests	Extension of empirical defect strength criterion of Barton (1973). Parameters were determined based on "trial and error fitting of numerous triaxial data" Undefined at $\sigma_n = 0$
4.15	Charles (1991)	$\phi' = C_1\log(C_2/\sigma_3') + \phi_b'$	ϕ' - arctan τ/σ_n C_1, C_2 – constants ϕ_b - basic friction angle	Similar approach to Barton & Kjærnsli (1981) for principal stresses.
4.16	Gonzalez (1985)	$\phi' = \phi_0 - j\log_{10}\left(\frac{\sigma_{nf}}{\sigma_{c0}}\right)$	ϕ' - arctan τ/σ_n ϕ_0, j - see Figure 4.16 σ_{c0} - standard crushing grain strength	Logarithmic strength result based on results from literature. Note that the parameters require numerous large scale shear tests to be performed.

σ_n = Normal stress
σ_c = Unconfined compressive strength of the intact rock pieces
c_u = Coefficient of uniformity
γ = Unit weight
d_{50} = Particle diameter at which 50% of the material is finer

$$\phi' = \phi_0 - j\log_{10}\left(\frac{\sigma_{nf}}{\sigma_{c0}}\right)$$

$$\phi_0 = \phi_A + j\log_{10}\sigma_{c0}$$

$$j = j_0 - \Omega\log_{10} N_R$$

where,

j_0 = experimental parameters, constant for specific material and condition

Ω = experimental parameters, constant for specific material and condition

$$N_R = \left(\frac{\pi_4}{\pi_7}\right)^{1/3}\pi_3\pi_5^2$$

where,

π_3 = coefficient of uniformity $= C_u$

π_4 = crushing strength number $= S_{en}\sigma_{cm}$

π_5 = volume concentration of solids $= V_S/V_T = 1/(1+e)$

π_7 = shape factor $= 1/r_v$

$$N_R = \left(\frac{S_{en}\sigma_{cm}}{r_v}\right)^{1/3} C_u\left(\frac{1}{1+e}\right)^2$$

$$S_{en} = \frac{6}{\gamma_s}\sum_{k=1}^{e}\frac{\Delta P_k}{\bar{d}_k}$$

$$\sigma_{cm} = \frac{P_{d_{\max}}}{d_{\max}^2}$$

V_S = Volume of solids

V_T = Total volume

e = void ratio

γ_s = particle dry density $= G_s\gamma_w$

ΔP_k = percentage of material of fraction k with mean size d_k

$\bar{d}_k$ = mean particle diameter for fraction $k = \sqrt{d_{j+1}d_j}$

d_j = article size of sieve j

P_{dmax} = crushing load of particle of nominal size d_{max}

d_{max} = maximum particle size

Figure 4.16. Gonzalez (1985) shear strength of rockfill equations

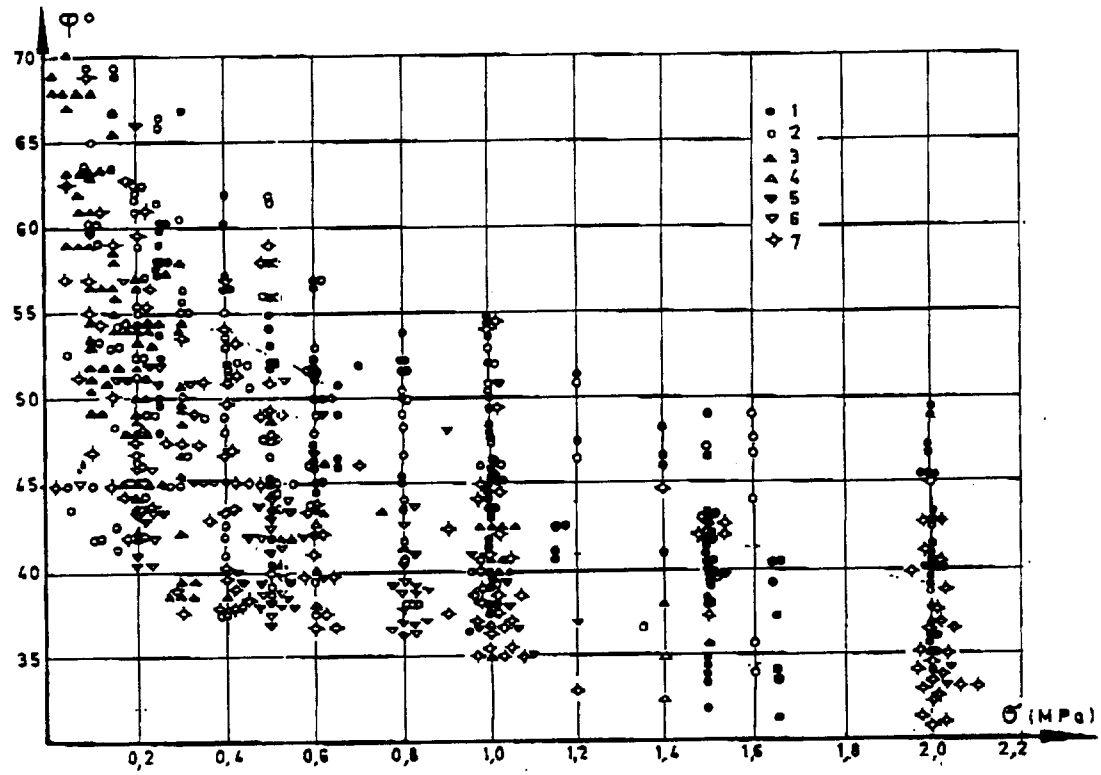


Figure 4.17. Friction angle, ϕ, vs normal stress, σ_n.

The constant B reflects curvature and, based on the direct shear tests, varies between 0.419 and 0.911 (Figure 4.18). It was found that B was lower for quarry materials than gravels (approx 0.67 c.f. 0.87 from). Lower B values were found to occur for material with lower particle strength, uniform grain size distribution and higher density. The constant A varies with similar factors to B and was found to increase with density (Figure 4.19) and mean grain diameter (Figure 4.20).

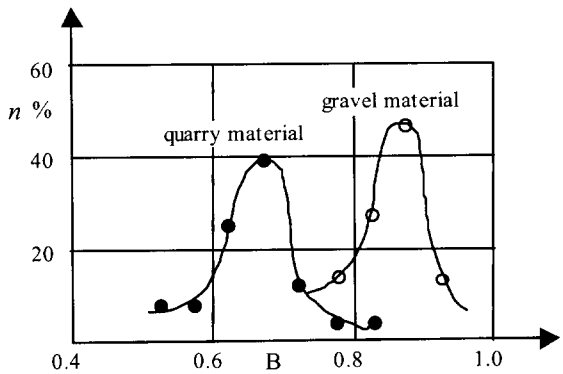


Figure 4.18. Frequency distributions for coefficients B versus porosity (Sarac & Popovic, 1985)

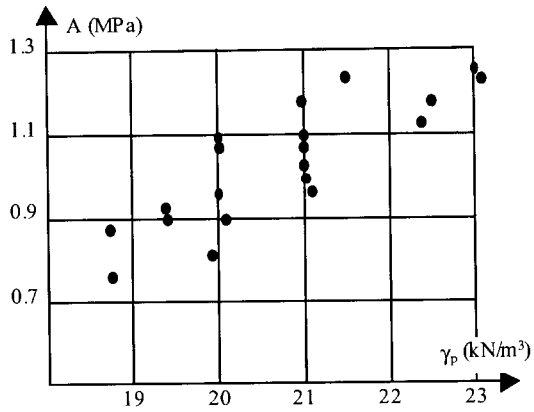


Figure 4.19. Dependence between coefficient *A* and gravel unit weight (Sarac & Popovic, 1985)

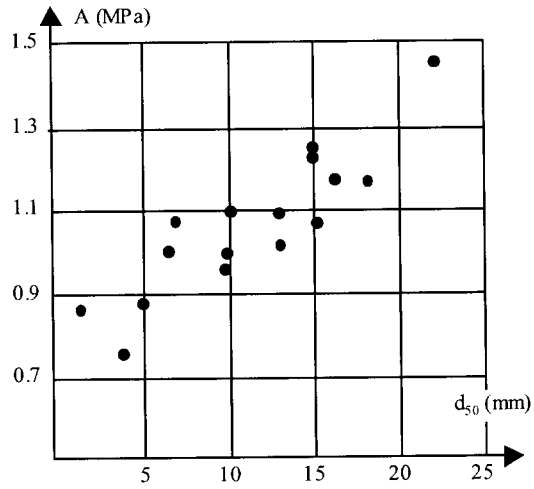


Figure 4.20. Dependence between coefficient *A* and mean grain diameter, d_{50} (Sarac & Popovic, 1985)

Barton & Kjærnsli (1981) Developed a rockfill shear strength criterion, Equation 4.14, based on the empirical joint shear strength criterion developed by Barton (1973). This equation has a dilation component dependent on equivalent strength and normal stress.

Based on "trial and error fitting of numerous triaxial data", Barton & Kjærnsli (1981) developed methods for estimating *S* based on the uniaxial compressive strength, σ_c, and d_{50}. The d_{50} was chosen as a reasonable descriptor of the rockfill. Barton & Kjærnsli (1981) claim that variations in rockfill strengths due to different gradings (wide or narrow) are due to porosity differences and not *S*. Barton & Kjærnsli (1981) present curves for triaxial and plain strain strengths to account for the higher ϕ' values normally obtained in plane strain tests. The reduction is incorporated in *S* as the ϕ' reduction is

assumed to be caused by less crushing of particles in the plain strain test. The equivalent roughness, R, is obtained using porosity and particle shape. ϕ_b is estimated indirectly using tilt tests on dry, flat, non-dilatant samples of rock surfaces (joints).

Gonzalez (1985) develops a logarithmic strength envelope (Equation 4.16) from an analysis of work by others. The equations used to develop the strength envelope are given in Figure 4.16. This procedure requires a large number of large-scale triaxial tests to provide the input data. Charles (1991) assumes that $c = 0$ for each test and therefore presents Equation 4.15, which is of similar form to Equation 4.14.

The equations in Table 4.5 are empirically derived. As such, they can be expected to be reasonable for the rockfill types and stress ranges for which they were created. Extrapolation beyond the bounds of the data used to create the criteria is cautioned against.

4.4 DATABASE OF TRIAXIAL SHEAR TESTS

The author has collated a database of large-scale triaxial shear tests. The database has over 989 individual triaxial shear tests from 307 sets of data. The information in the database has been collected from selected quality published papers and reports and from unpublished laboratory reports from organisations that have carried out high quality testing for dams and other projects. There is an assumption inherent in collating a large amount of data that all data sets are of equal quality. The dangers of making this assumption were mitigated to some degree by using the quality sources stated above. Although virtually all tests were accepted, there were some that were rejected. Tests that were carried out at a very high strain rate were rejected. As were results where the strength of the sample was estimated due to the test reaching the strain limit of the testing apparatus prior to attaining the maximum strength of the sample. Table 4.6 shows the parameters recorded in the database. The complete database is contained in Appendix F on a CD-ROM that is appended to this thesis.

Table 4.7 summarises the basic statistics of the data. The triaxial cells used for tests ranged from a diameter of 50.8mm to a diameter of 1130mm. The maximum particle size in the tests ranged from 4.8mm to 200mm. The minimum particle size in the tests ranged from 0.0035mm to 40mm. Note that many tests recorded a minimum particle size of 'less than' a particular diameter.

40% of tests had zero fines (% passing 0.075mm) content whilst 80% of the tests had a fines content of no more than 5%. 10% of the triaxial tests had fines contents in excess of 20%. The largest fines content for a test in the database was 45%. Note that the uniformity coefficient and the coefficient of curvature are very high for tests where fines have been added to a large diameter rockfill.

18.7% of the triaxial tests were on basalt material whilst 14.9% were on granite, 8.2% sandstone and 3.9% limestone. Many tests did not give the rock type of the material tested.

49% of the samples tested were considered angular (with an angularity rating = 8).

Table 4.6. List of parameters in triaxial shear strength database

Parameter	Description
Case	Test number within database
Reference	Data source
Data set	Identifying number given for each set of tests performed. Taken from reference where given.
Test	Test number within data set
Site	Place of origin of material or location (e.g. dam name)
Origin	Source of gravel size particles (e.g. crushed, blasted, alluvial etc.)
Type	Rock particle type (e.g. sandstone etc)
UCS (MPa)	Unconfined compressive strength of individual particles
UCS* (MPa)	Includes UCS estimated where not given (see Section 4.5.1)
Ang	Angularity of particles from: angular; sub-angular; sub-rounded; rounded
Ang rat	A value from 1 (rounded) to 8 (angular) based on a description of the rockfill given in reference. (0 = not given)
Ds (mm)	Sample diameter
Hs (mm)	Sample height
Fines (%)	Percent of material finer than 0.075mm
Fines* (%)	Includes estimated fines where not given (see Section 4.5.1)
d_{min} (mm)	Minimum particle size
d_{10} (mm)	Diameter corresponding to 10% finer
d_{30} (mm)	Diameter corresponding to 30% finer
d_{50} (mm)	Diameter corresponding to 50% finer
d_{60} (mm)	Diameter corresponding to 60% finer
d_{90} (mm)	Diameter corresponding to 90% finer
d_{max} (mm)	Maximum particle size
c_u	Coefficient of uniformity: $c_u = \frac{d_{60}}{d_{10}}$
c_c	Coefficient of curvature: $c_c = \frac{d_{30}^2}{d_{10} d_{60}}$
Grav (%)	Gravel content (defined as percent of material greater than 2mm diameter)
r_d	Relative density
g_d (kN/m^3)	Dry unit weight
e_{min}	Minimum void ratio
e_{max}	Maximum void ratio
e_i	Initial void ratio
e_i^*	Includes estimated void ratio where not given (see Section 4.5.1)
n	Porosity
B_g	Breakage co-efficient (Marsal)
G_S	Specific gravity
w (%)	Moisture content
S (%)	Degree of saturation
σ'_3	Minimum principal effective stress
σ'_1	Maximum principal effective stress
ε_{af}	Axial strain at failure
ε_{vf}	Volumetric strain at failure
Brit	Type of failure (B-brittle, T-transitional, D-ductile). Qualitative term based on the stress-strain curves (if given) presented in the reference.
f_{sec}	Secant friction angle (calculated)
σ_n	Normal stress on failure plane (calculated)
σ_1-σ_3	Principal stress difference (calculated)
σ_1/σ_3	Principal stress ratio (calculated)

Table 4.7. Summary of basic statistics from the rockfill database

Parameter	Minimum	Average	Maximum
Sample diameter (mm)	50.8	332	1130.3
Minimum particle size (mm)	0.0035	3.6	40
d_{10} (mm)	0.001	4.3	53
d_{30} (mm)	0.01	9	90
d_{50} (mm)	0.09	16	100
d_{60} (mm)	0.22	20	110
Maximum particle size (mm)	4.8	59	200
Fines content (%)	0	4.1	45
Uniformity coefficient, c_u	1.3	81	3243
Coefficient of curvature, c_c	0.04	2	16.9
Unconfined compressive strength of intact rock particles (MPa)	25	153	761

4.5 DATABASE ANALYSIS

This section describes the qualitative and quantitative (using statistical methods) analysis of the rockfill database. Analyses using both the secant friction angle and principal stresses were carried out. A final analysis using the Hoek-Brown criterion was performed as part of the development of a criterion for rock mass strength.

4.5.1 Analysis Methodology

The database contained missing information for some of the test results. If a parameter used in the statistical analysis was missing for a particular test, that test result was generally ignored. However, to increase the number of tests, certain assumptions were made prior to the analysis. These are:

Where an unconfined compressive strength of the intact rock was not given and reasonable estimates could be made, it was estimated into three groupings of 40, 100 and 200MPa rocks based on a description of the rock type, degree of weathering and also the origin of the material. These samples were placed. The effect of including these data sets was evaluated during the statistical analysis.

Where the minimum particle size, d_{min}, was recorded as 'less than' a particular diameter the following table was used to estimate d_{min} for the analysis.

Minimum particle size given, d_{min}	Minimum particle size used, d_{minhat}
<0.075 (and d_{10}>0.075)	0.01
<0.075 (and $d_{10}\leq$0.075)	0.005
<0.02	0.005
<0.005 (and d_{10}>0.003)	0.001
<0.005 (and $d_{10}\leq$0.003)	0.0005

Where an initial void ratio was not given it was calculated from one of the following equations (in order of preference):

$$e = \frac{n}{1-n} \tag{4.17}$$

$$e = e_{\max} - D_r \left(e_{\max} - e_{\min} \right) \tag{4.18}$$

$$e = \frac{G_s g_w}{g_d} - 1 \tag{4.19}$$

where

e = void ratio

n = porosity

e_{max} = maximum void ratio

e_{min} = minimum void ratio

D_r = relative density (%)

g_w = unit weight of water (kN/m^3)

g_d = dry unit weight (kN/m^3)

G_s = specific gravity (assumed = 2.7 if not given)

4.5.2 Secant Friction Angle, ϕ_{sec}, Versus Normal Stress, σ_n

Relationships between secant friction angle, ϕ_{sec}, and normal stress, σ_n, were investigated using the rockfill shear strength database. This relationship was chosen as it is commonly used in the literature and practice to present and analyse data. The results are discussed in detail below however, it should be noted at the outset that the fits obtained using this statistical analysis were inferior to those obtained using a relationship of maximum principal stress, σ'_1, versus minor principal stress, σ'_3. This is of particular importance at both low and high confining stress, σ'_3.

4.5.2.1 General Assessment of Database

Figure 4.21 to Figure 4.27 show plots of ϕ_{sec} vs σ_n with σ_n plotted on a logarithmic scale. These plots are discussed in turn below. Note that these analyses were carried out for the complete data sets. No parameter (e.g. σ_c) was estimated for these plots. Where a specific parameter has been investigated a small '+' symbol on a figure represents a test result where the parameter is unknown (except for Figure 4.21).

- *Figure 4.21 f_{sec} vs s_n*: This figure shows that ϕ_{sec} has a range of approximately 25° for any given value of σ_n. This shows that the bounds given by Indraratna et al (1993) (equating to a range of about 15°) for this type of plot do not indicate the full range of ϕ_{sec} possible. Their upper bound appears to be too conservative whilst the lower bound, although providing a better fit, has several test results below it. The gradient of the bounds does not appear to be steep enough to adequately describe the variation in ϕ_{sec} relative to σ_n.

- *Figure 4.22 f_{sec} vs s_n sorted on angularity rating*: The addition of angularity rating does not assist greatly with improving the prediction of ϕ_{sec} vs σ_n. The figure shows that angular rockfill (angrat = 7 or 8) generally lies above sub-angular material (angrat = 5 or 6). However, rounded material (angrat = 1 or 2) lies roughly in the centre of the data with angular material lying both above and below the data.

- *Figure 4.23 f_{sec} vs s_n sorted on rock type = basalt*: This was plotted to see the amount of variability there was within one class of rock type. Basalt is used as railway ballast, road base and dam rockfill. A basalt rockfill might be expected to lie at the higher strength end of all rockfills. Although the figure shows this to be true there is still a 10°-15° range in ϕ_{sec} for the material for any given σ_n.

- *Figure 4.24 f_{sec} vs s_n sorted on coefficient of uniformity, c_u*: The figure shows that c_u does not appear to be a very good additional predictor with data for each c_u range generally being well and evenly spread. It could be argued that the lower strength rockfills predominately have high c_u values (>12). However, these types of rockfills are spread throughout the full strength range.

- *Figure 4.25 f_{sec} vs s_n sorted on maximum particle size, d_{max}*: This data is also quite scattered although there appears to be a tendancy for high d_{max} rockfills to result in a higher strength than small d_{max} rockfills.

- *Figure 4.26 f_{sec} vs s_n sorted on percent fines content*: The rockfills with a high fines content (>20%) generally have a lower ϕ_{sec}. Much of the other data, particularly with fines less than 10% is well scattered.

- *Figure 4.27 f_{sec} vs s_n sorted on unconfined compressive strength, s_c*: Rockfills with a σ_c of less than 100MPa appear to have lower strengths than those greater than 100MPa. It is difficult to discern any real difference in the results for rockfills with strengths of 100-200MPa and greater than 200MPa.

A general conclusion from the analysis of the graphs may be made that the factors described above do have some effect on the shear strength of rockfill, although none of them stands out as being substantially better than the others as a predictor. It is unwise to estimate strength based solely on these graphs due to the interrelationship between the various factors in the tests. A better statistical analysis is to perform a global analysis of the data using all these factors and assessing whether they are statistically significant. This is performed in the next section.

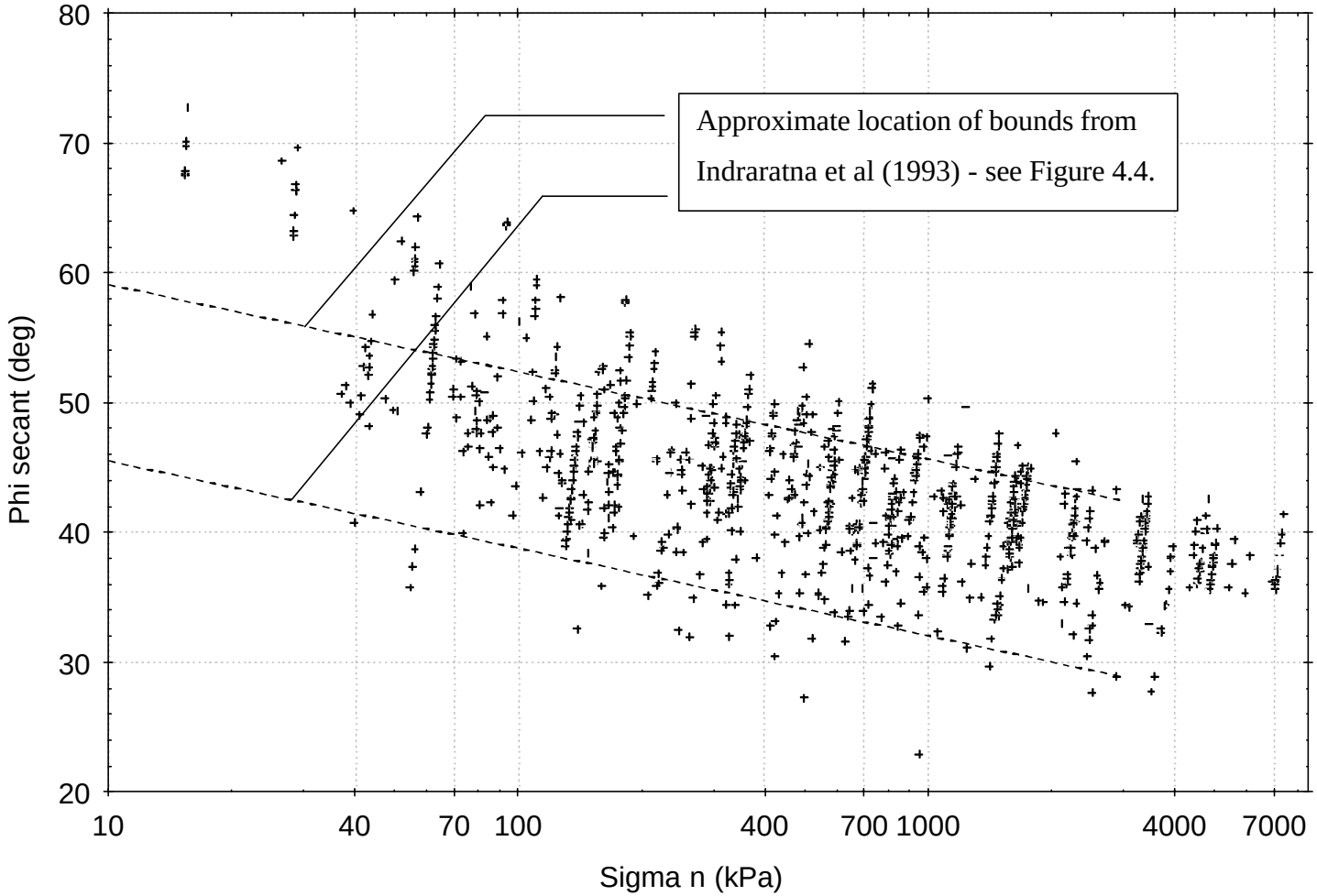


Figure 4.21. Secant friction angle, f_{sec} vs normal stress, s_n

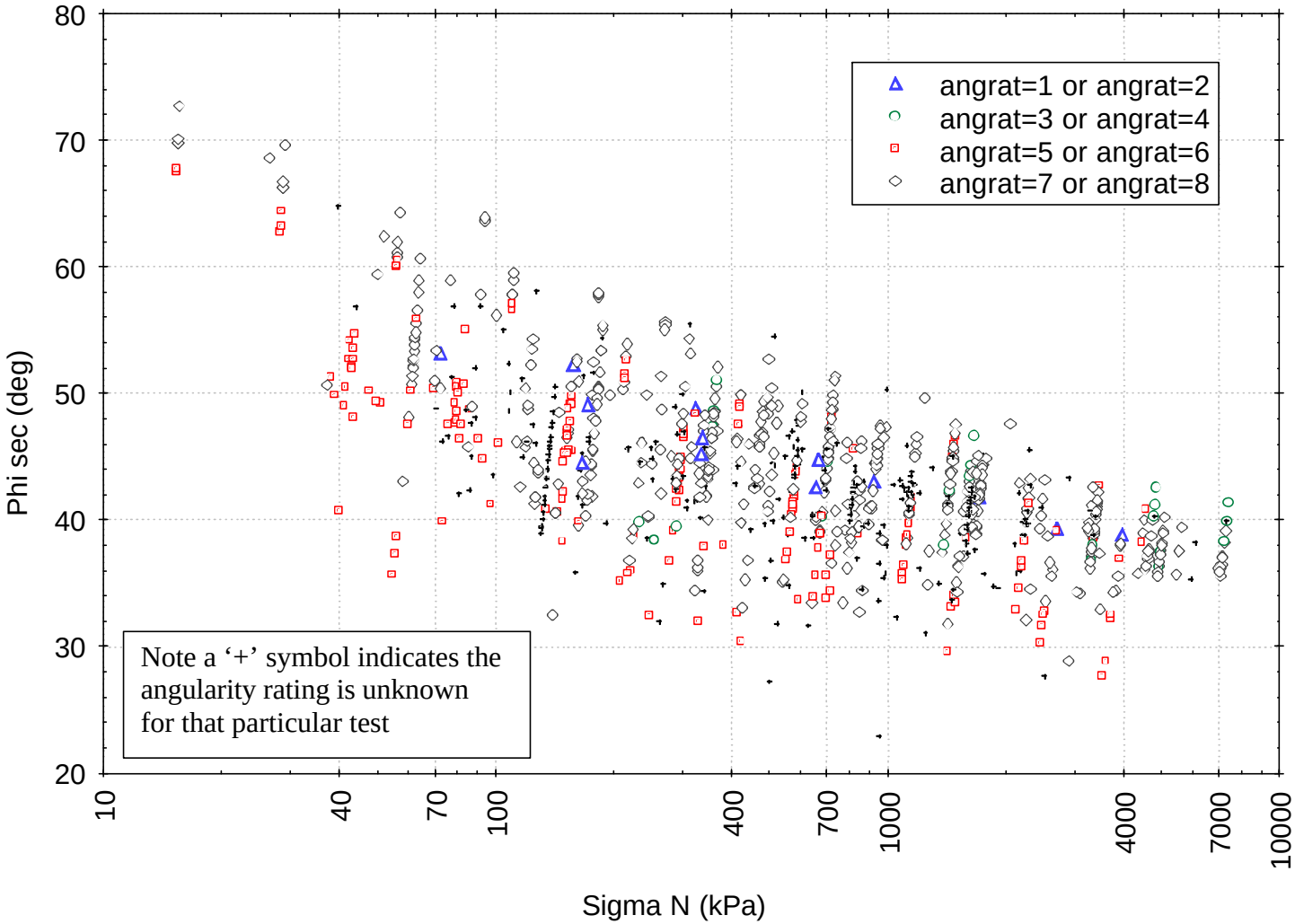


Figure 4.22. Secant friction angle, f_{sec} vs normal stress, s_n, sorted on angularity rating

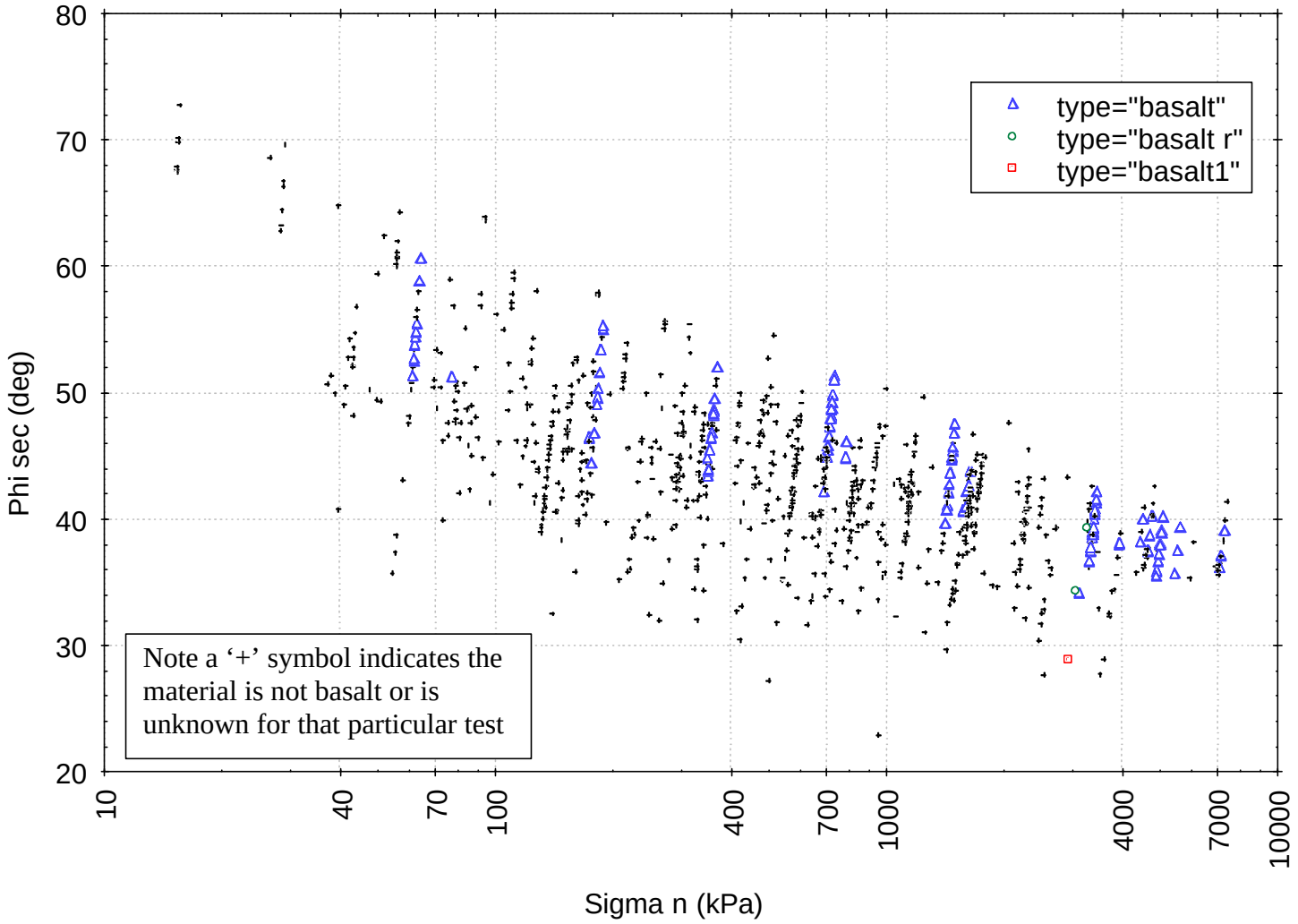


Figure 4.23. Secant friction angle, f_{sec} vs normal stress, s_n, sorted on rock type = basalt

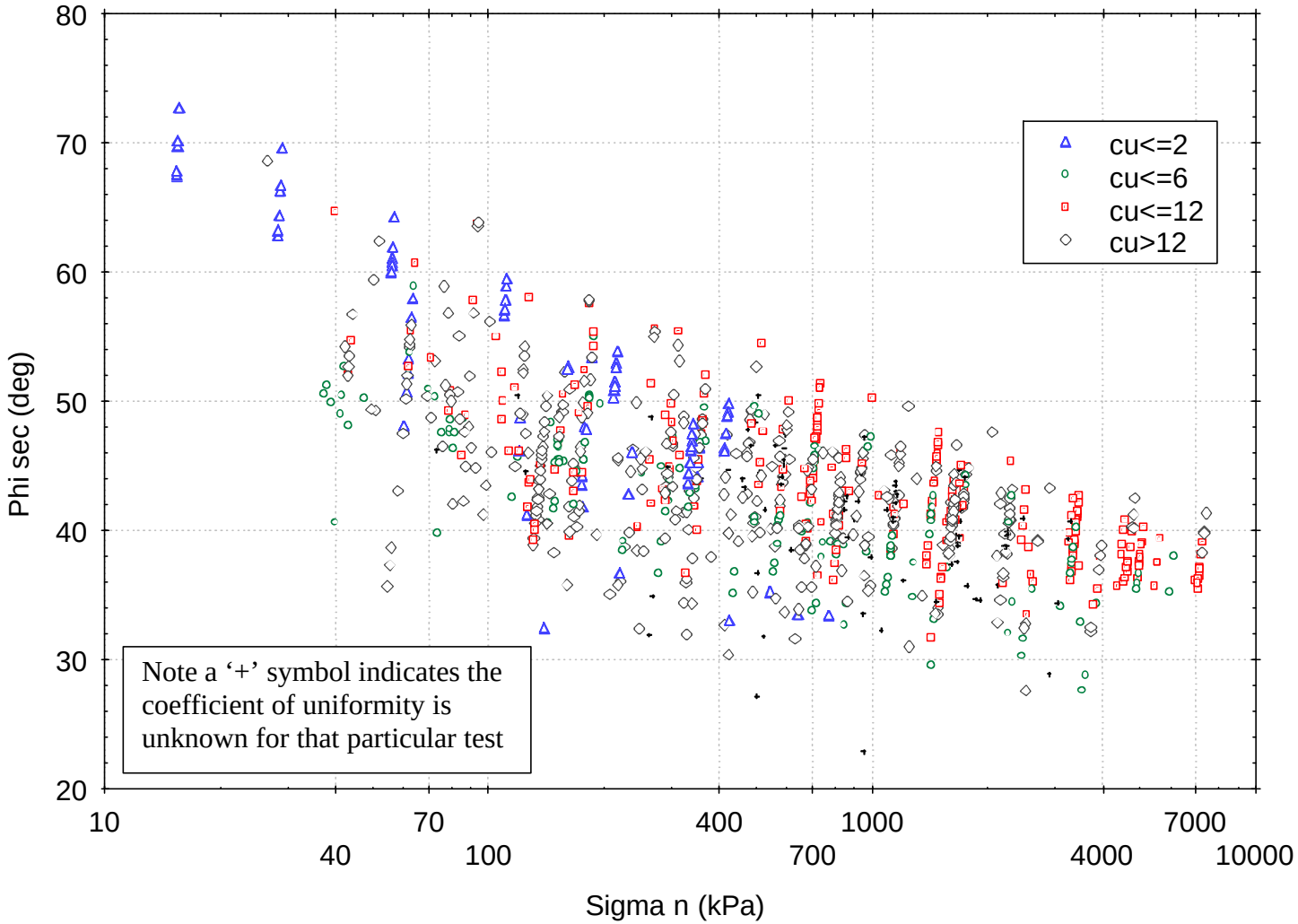


Figure 4.24. Secant friction angle, f_{sec} vs normal stress, s_n, sorted on coefficient of uniformity, c_u

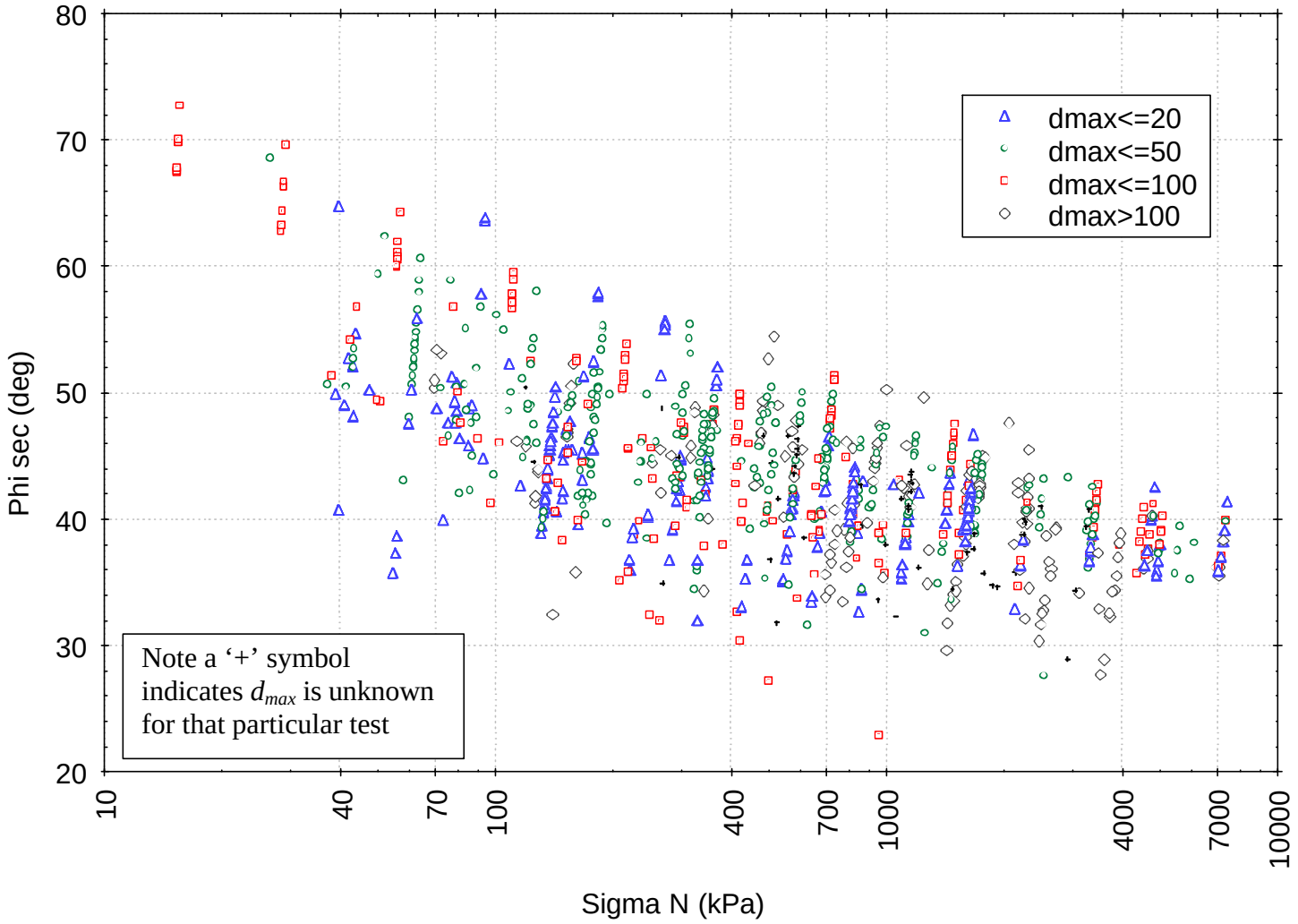


Figure 4.25. Secant friction angle, ϕ_{sec} vs normal stress, σ_n, sorted on maximum particle size, d_{max}

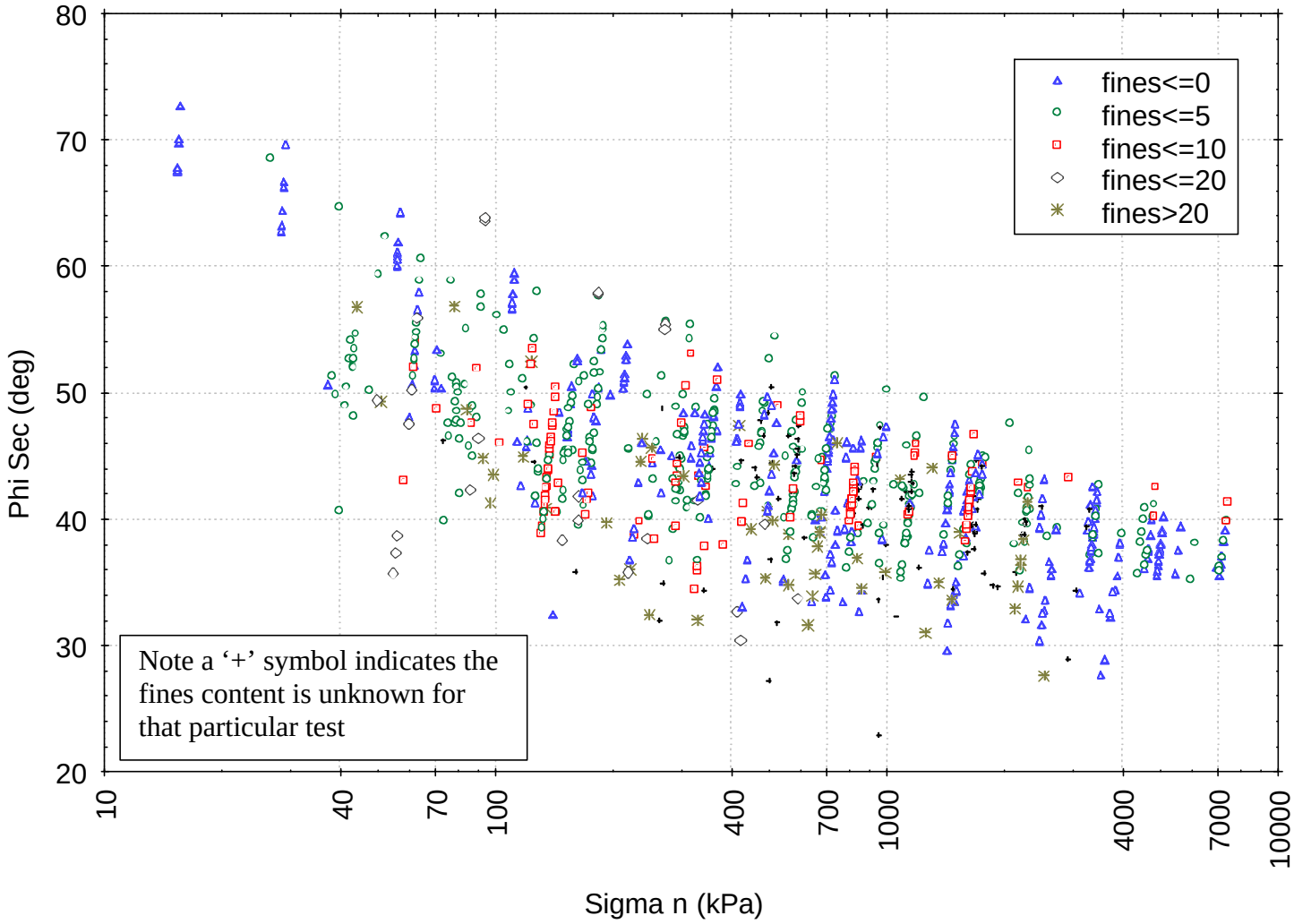


Figure 4.26. Secant friction angle, f_{sec} vs normal stress, s_n, sorted on percent fines (passing 0.075mm) content

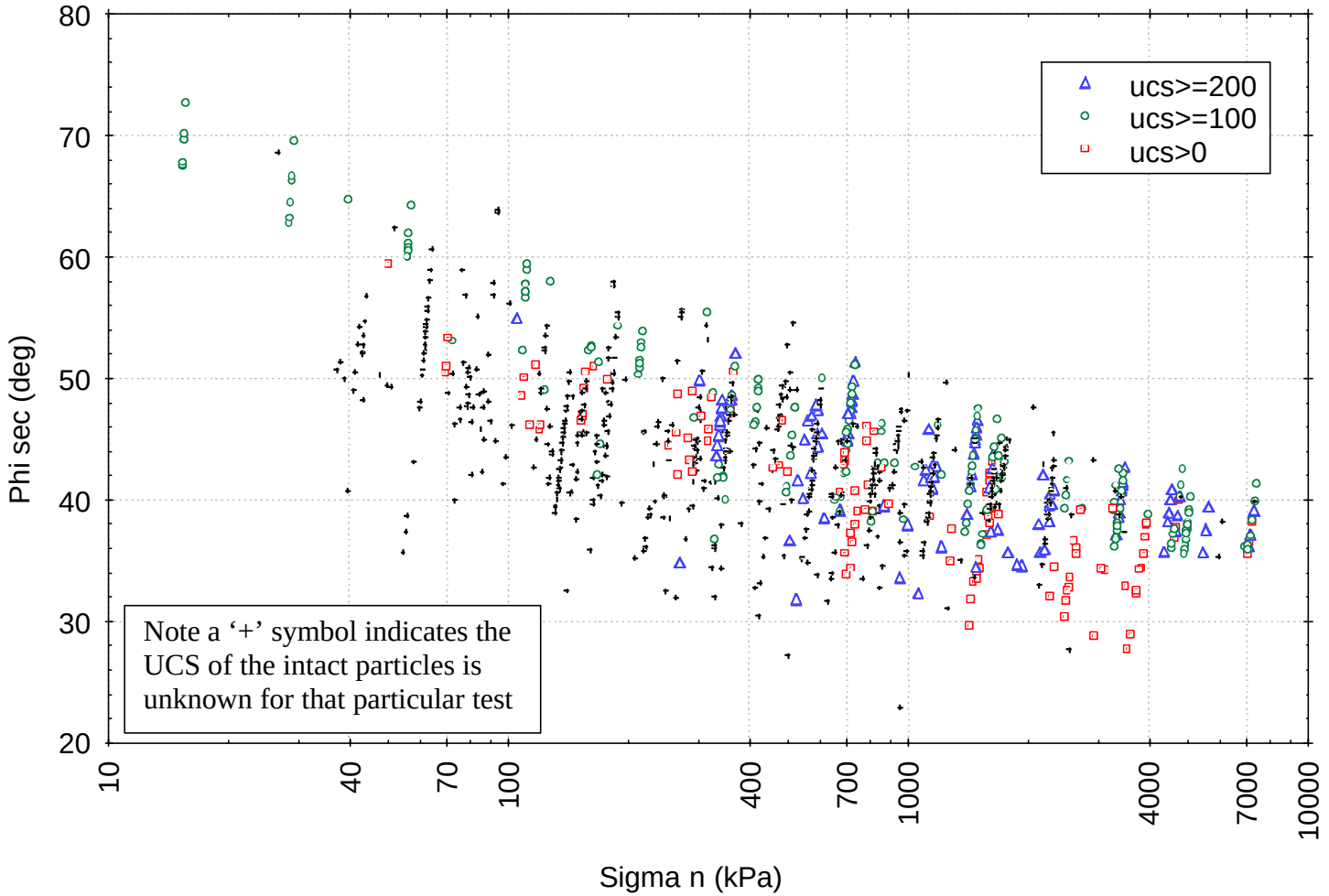


Figure 4.27. Secant friction angle, f_{sec} vs normal stress, s_n, sorted on unconfined compressive strength of the rock substance, UCS (MPa)

4.5.2.2 Statistical Analysis of Database

A non-linear statistical analysis was carried out on the database. The estimation method used was least squares. An analysis of the form shown below was found to be the most effective for relating ϕ to σ_n.

$$f' = a + b s_n'^c \tag{4.20}$$

The analysis resulted in the following constants:

$$a = 36.43 - 0.267\,ANG - 0.172\,FINES + 0.756(c_c - 2) + 0.0459(UCS - 150) \tag{4.21}$$

$$b = 69.51 + 10.27\,ANG + 0.549\,FINES - 5.105(c_c - 2) - 0.408(UCS - 150) \text{-0.408} \tag{4.22}$$

$$c = -0.3974 \tag{4.23}$$

where,

$ANG = (angularity\,rating - 5)$ for angularity rating >5.5; otherwise 0

FINES = percentage of fines passing 0.075mm (%)

c_c = coefficient of curvature

UCS = unconfined compressive strength of the rock substance (MPa)

This function resulted in a variance explained of just 61.7%. It should be noted that the addition of the uniformity coefficient, c_u, and the maximum and minimum particle sizes, d_{max} and d_{min}, did not result in a better fit to the data. As the ϕ vs σ_n curve is unconstrained at both low and high σ_n it is unlikely that a reasonable fit will be found regardless of the function used. Due to the relatively poor statistical results it was decided to proceed with an analysis using principal stresses.

4.5.3 Maximum Principal Stress, σ'_1, versus Minimum Principal Stress, σ'_3

The analysis using ϕ_{sec} vs σ_n gave poor results. In this section principal stresses are used. These are constrained at σ'_3=0, σ'_1=0. A non-linear statistical analysis was carried out on the database. The estimation method used was least squares. An analysis of the form shown in Equation 4.24 was found to be the most effective for relating σ'_1 to σ'_3.

$$\sigma'_1 = RFI\,\sigma'^{a}_3 \tag{4.24}$$

An initial analysis on the whole database (988 data sets) found (Variance explained = 97.13%):

RFI = 4.7002

α = 0.8972

To determine which parameters affected RFI, the analysis was carried out iteratively. The analysis was carried out using a parameter in the database as a predictor of RFI (e.g. σ_c). After each analysis the residuals were plotted against the parameters in the database. Where a trend was noted for a particular parameter another analysis was carried out using that additional parameter. The analysis resulted in the following equations (based on 869 data sets and with a variance explained = 98.82%):

$$a = 0.8726$$

$$RFI = 6.3491RFI_e + 0.48763RFI_{ANG} - 0.0027RFI_{d_{\max}} - 1.1568RFI_{FINES} + 0.30598RFI_{UCS} \tag{4.25}$$

$$RFI_e = \frac{1}{1+e_i} \tag{4.26}$$

$$RFI_{ANG} = 1 \text{ if angular, otherwise} = 0 \tag{4.27}$$

$$RFI_{d_{\max}} = d_{\max}\ (\text{mm}) \tag{4.28}$$

$$RFI_{FINES} = \frac{e^{(\text{Fines}-20)}}{1+e^{(\text{Fines}-20)}} \quad \text{where fines is in \%} \tag{4.29}$$

$$RFI_{UCS} = \frac{e^{(\text{UCS}-110)}}{1+e^{(\text{UCS}-110)}} \quad \text{where UCS is in MPa} \tag{4.30}$$

Equation 4.29 and 4.30 are effectively smooth "step" functions that are ameniable to equation solving routines.

Figure 4.28 shows the variation of RFI_e with initial void ratio. This shows that an increase in initial void ratio will lead to a decrease in σ'_1 and shear strength.

Highly angular rockfill (angularity rating = 8) showed an increase in strength. There was no noticeable trend for less angular rockfill (angularity rating ≤ 7) and hence Equation 4.27 was chosen.

The strength of the rockfill (σ'_1) was found to be proportional to the maximum particle size (Equation 4.28).

The unconfined compressive strength, σ_c, was found to have a limited effect on the strength of the rockfill. Generally strengths below a σ_c of 100MPa had similar strengths, as did those with σ_c greater than 120MPa. Equation 4.30 acts as a switch at a σ_c around 110MPa. Below a σ_c of 110MPa the function rapidly approaches zero and above a σ_c of 110MPa the function rapidly approaches unity (Figure 4.29).

The fines content was found to only affect the strength of the rockfill where the fines content was greater than approximately 20%. Equation 4.29 acts as a switch at a fines content around 20%. Below a fines content of 20% the function rapidly approaches zero and above a fines content of 20% the function rapidly approaches unity (Figure 4.30).

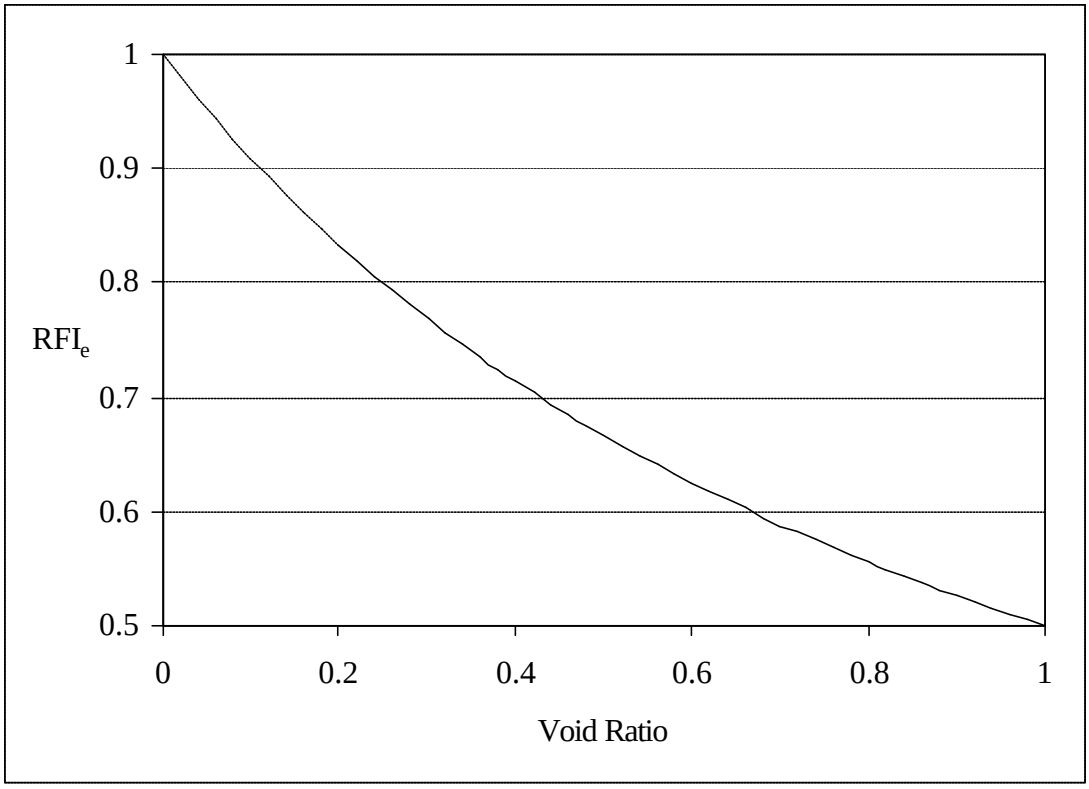


Figure 4.28. *RFI_e* versus void ratio

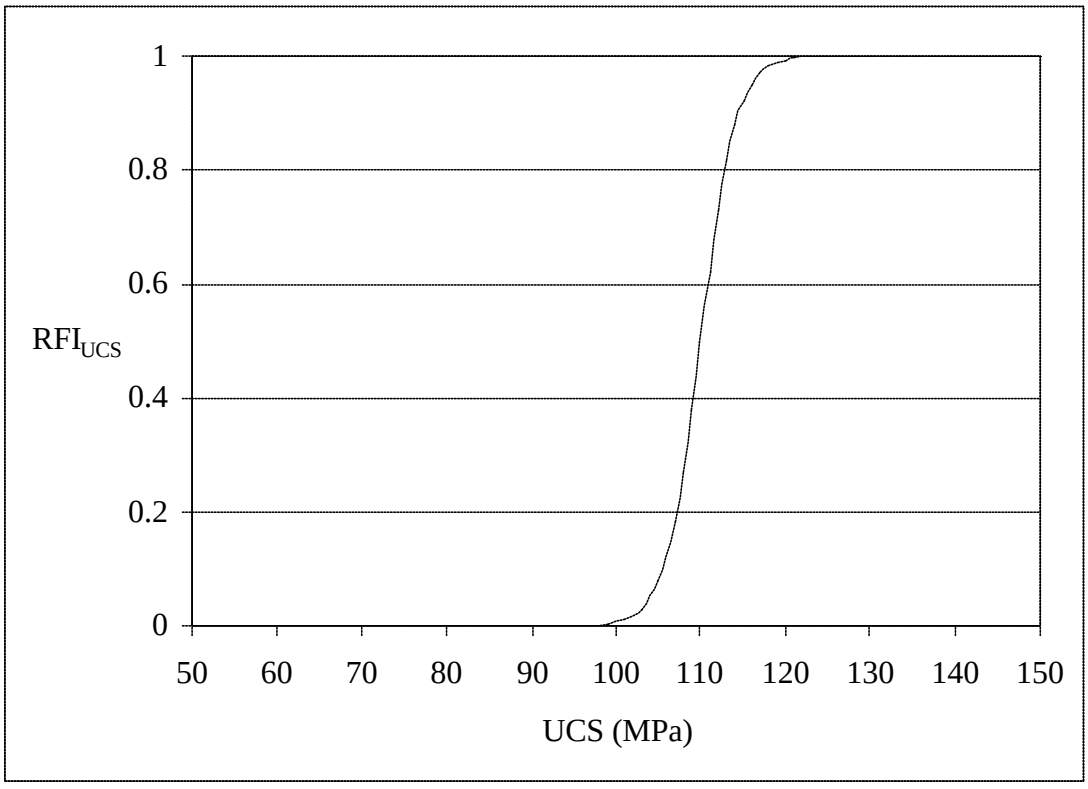


Figure 4.29. *RFI_{UCS}* versus unconfined compressive strength

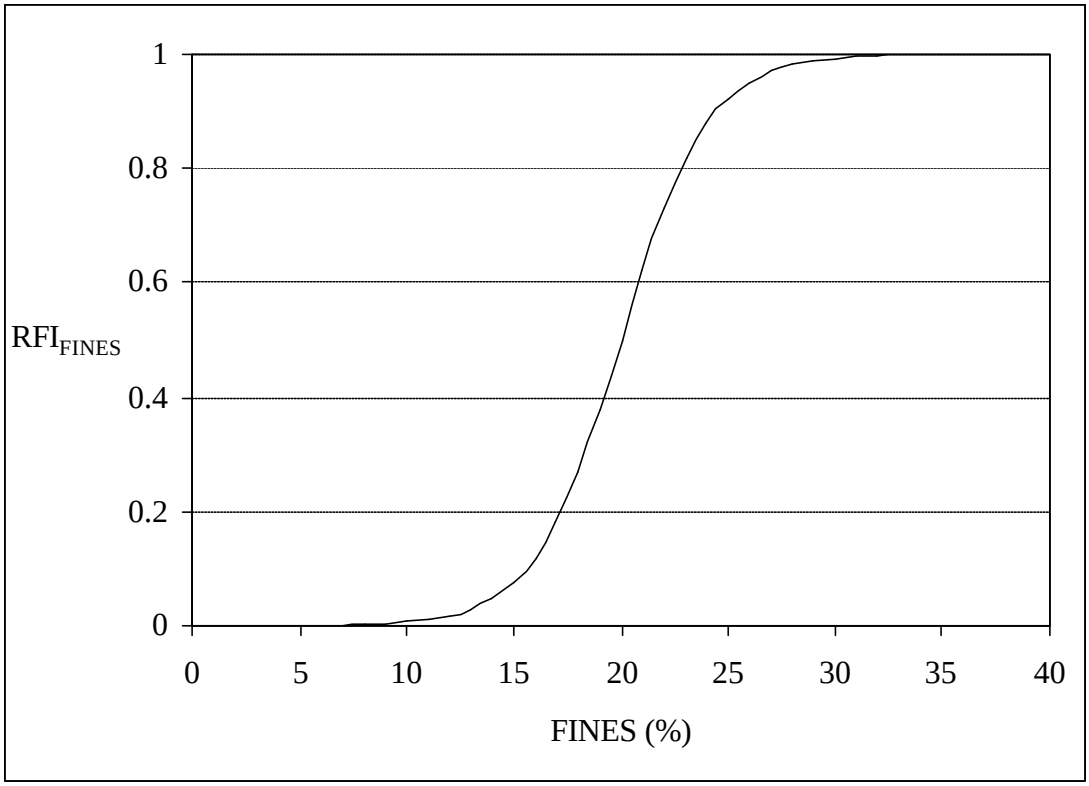


Figure 4.30. RFI_{FINES} versus percent fines

The residuals versus unconfined compressive strength, maximum particle size, initial void ratio, angularity and fines content are shown on Figure 4.31 to Figure 4.35 respectively. These figures shown no discernible trend from which it can be assumed that the equations provided above are reasonable over the range of the data.

Figure 4.36 shows the residuals plotted against sample diameter. The lack of any major trend in the data suggests that sample diameter does not have a large effect on rockfill strength at least up to diameters of about 1.2m. This suggests that the process of scaling using parallel grading lines when testing rockfill for use in the field is valid and should not affect the results. Note however, that if fines are increased due to the scaling strength (σ'_1) may be reduced.

Figure 4.37 to Figure 4.41 show the effect that a change in unconfined compressive strength, angularity, fines content, maximum particle size and initial void ratio have on σ'_1 respectively.

Figure 4.42 shows the data used in the analysis together with lines representing RFI=7 (approximate upper bound), RFI=4.7 (approximate best estimate) and RFI=3 (approximate lower bound). Figure 4.43 shows the same plot with all the data in the database plotted. It shows that the bounds are still reasonable and that the data left out of the analysis (due to insufficient information) would not have changed the results significantly. Figure 4.44 shows the same plot for a stress range (σ'_3) up to 1.5MPa.

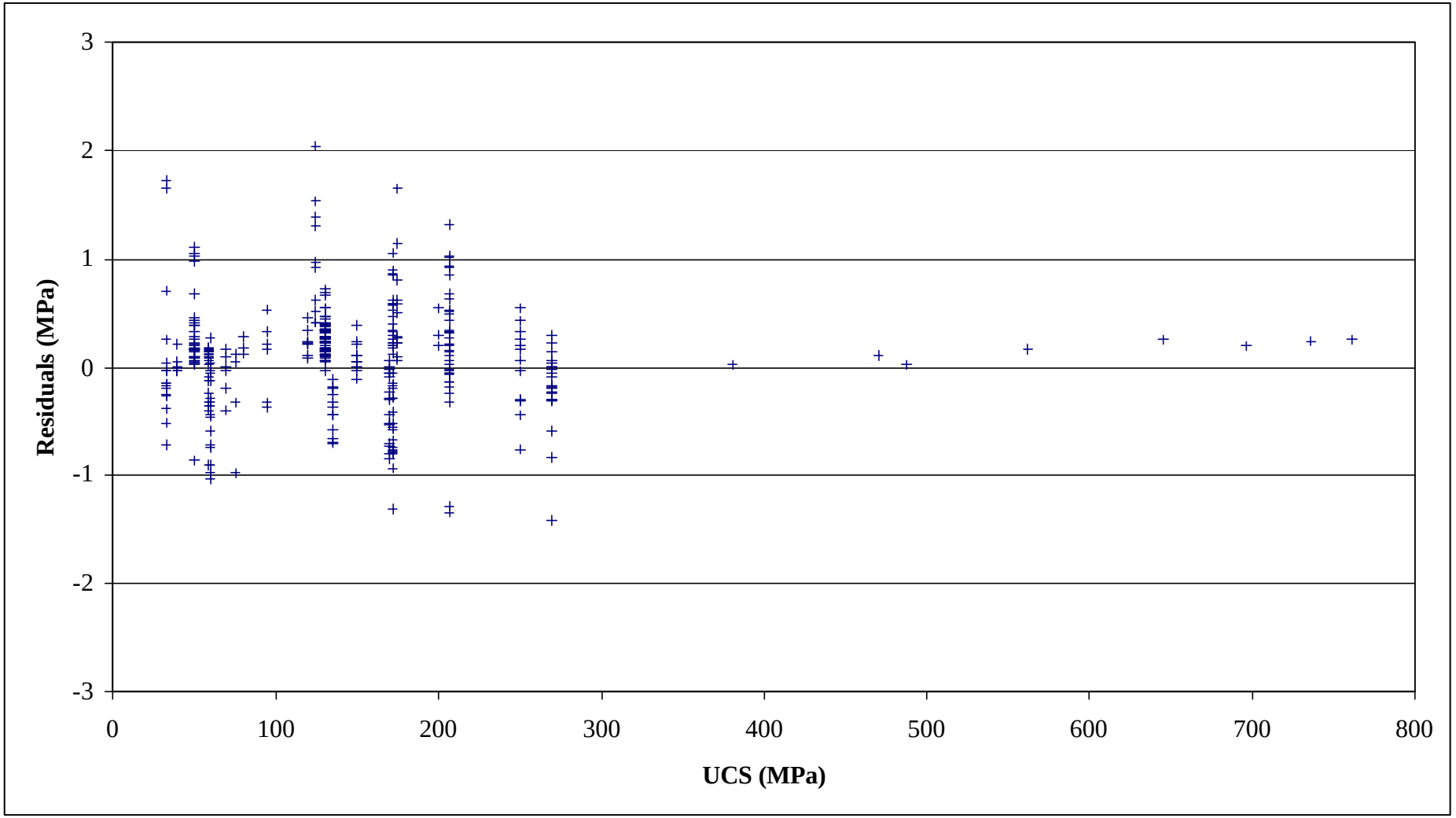


Figure 4.31 Residuals versus unconfined compressive strength of intact rock

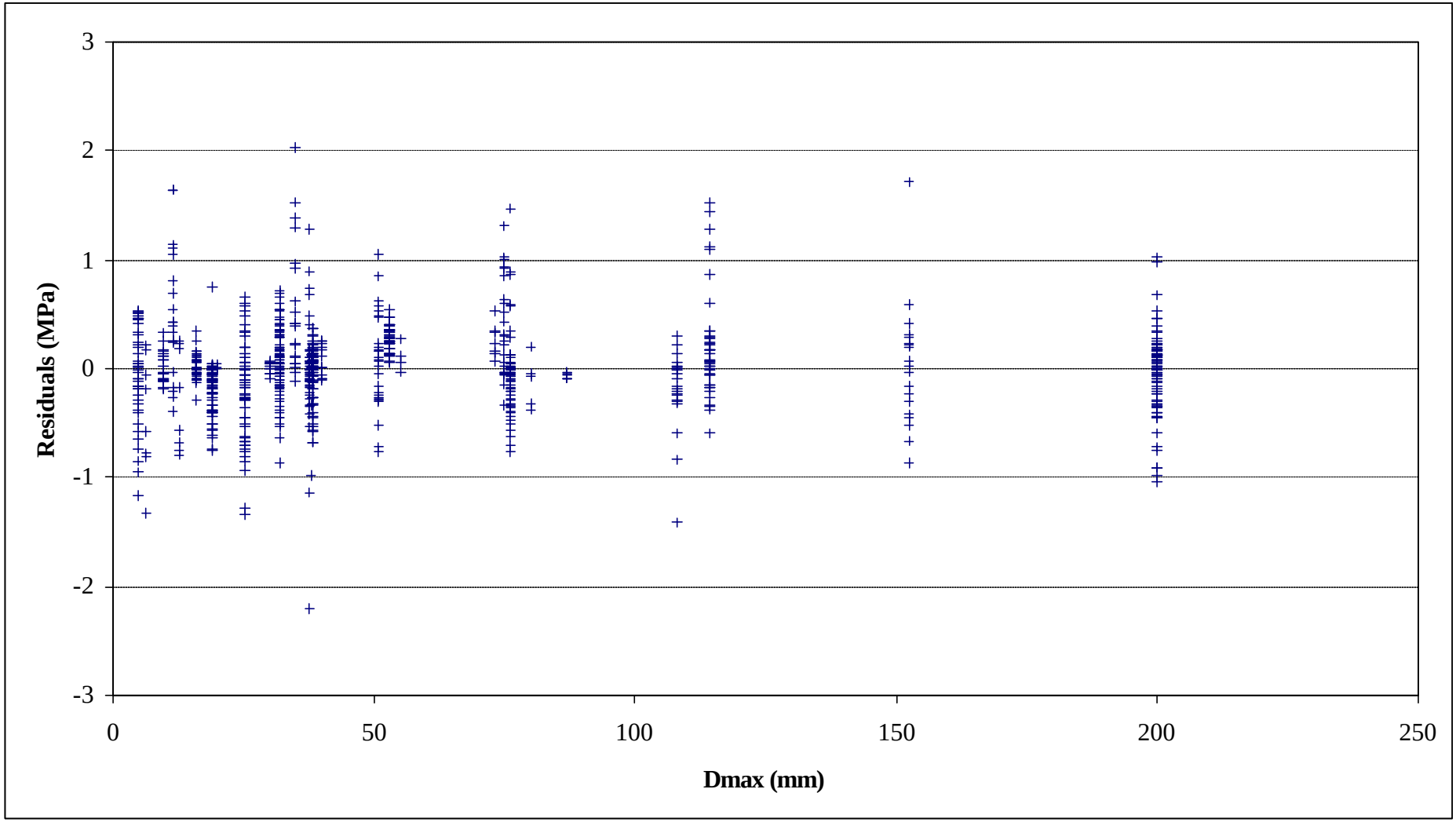


Figure 4.32. Residuals versus d_{max}

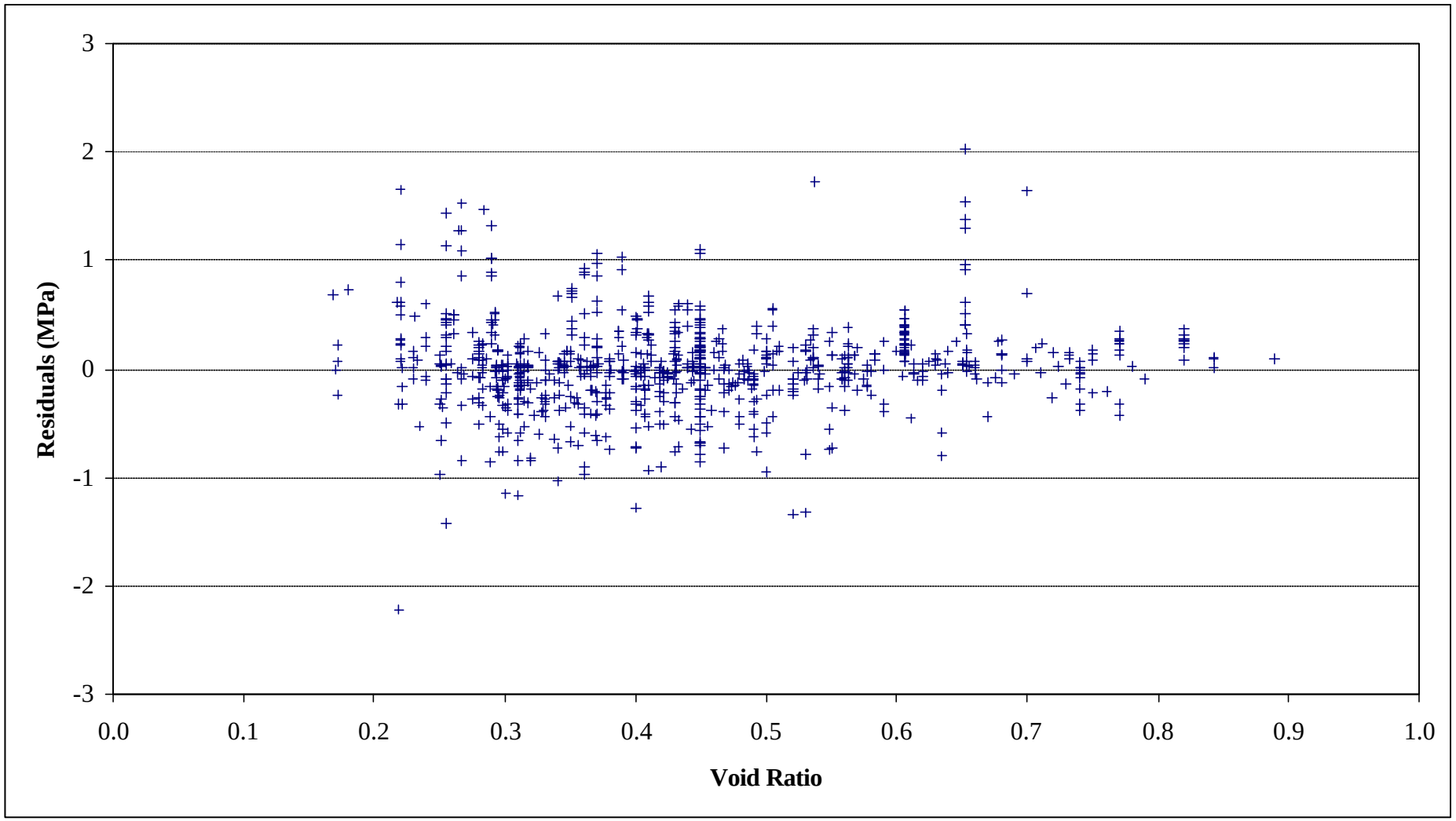


Figure 4.33. Residuals versus void ratio

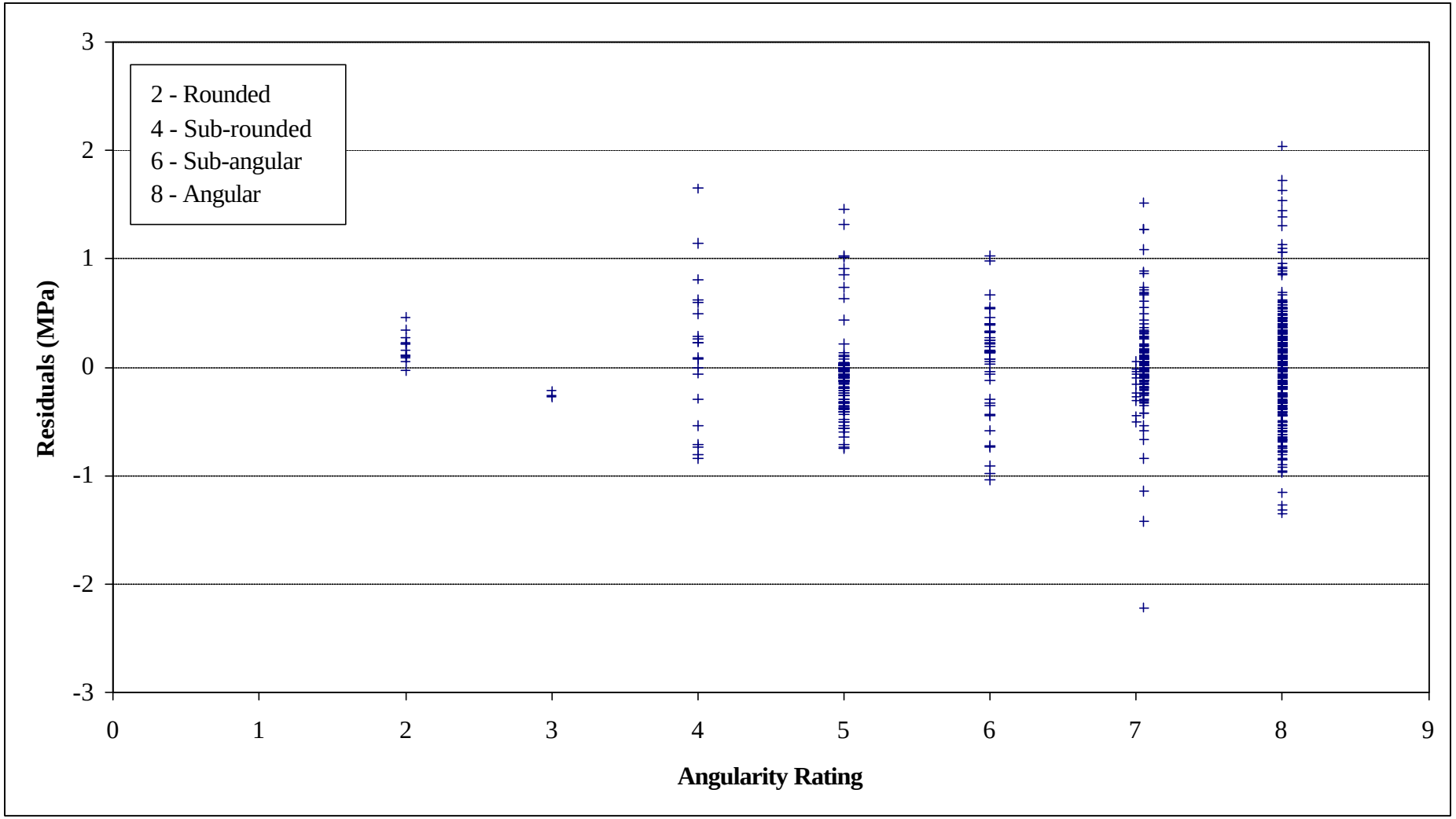


Figure 4.34. Residuals versus angularity rating

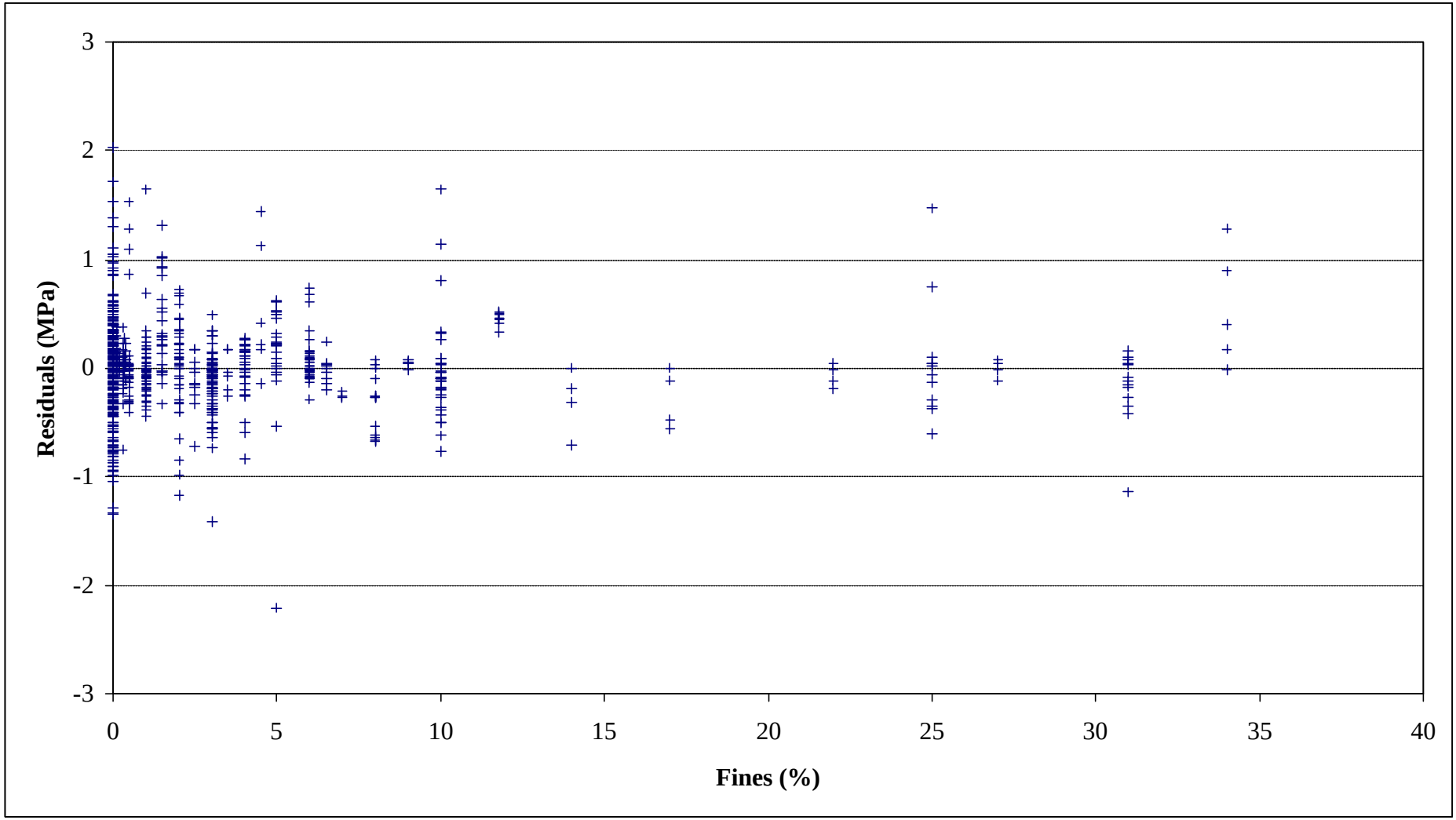


Figure 4.35. Residuals versus fines content

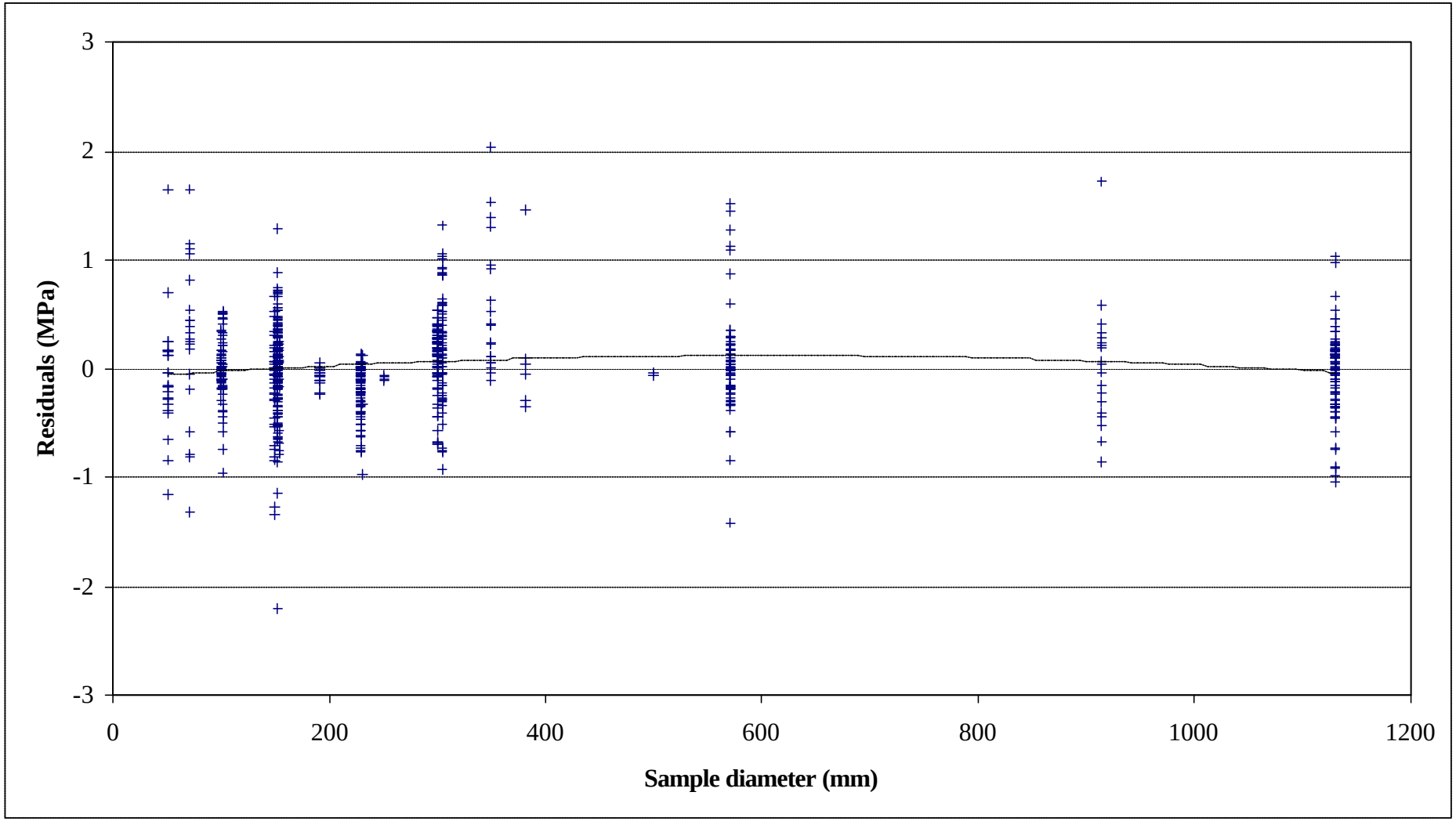


Figure 4.36. Residuals versus sample diameter

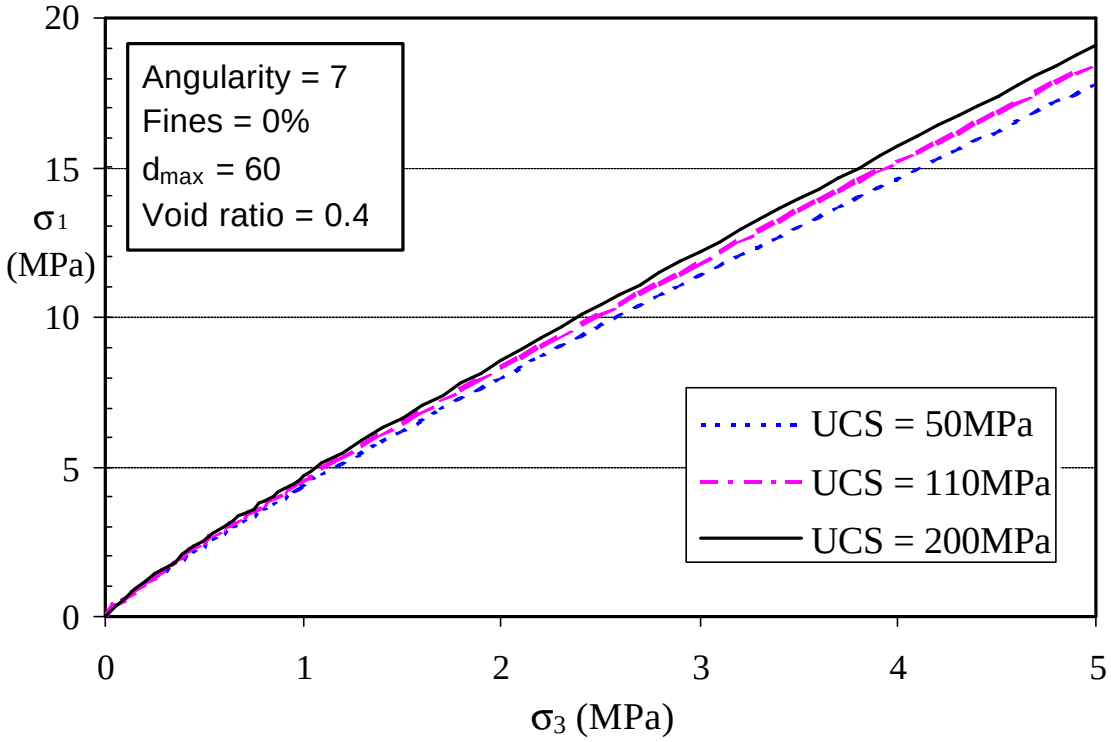


Figure 4.37. Effect of unconfined compressive strength on s_1

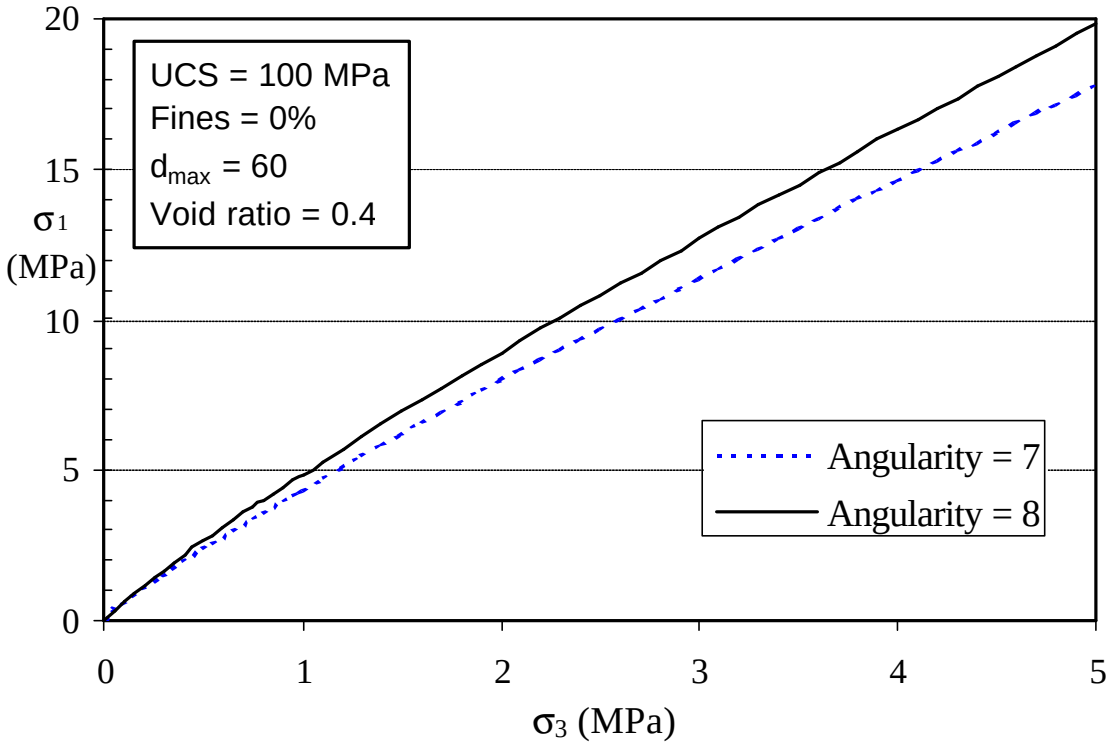


Figure 4.38. Effect of angularity on s_1 (7 = sub-angular to angular; 8 = angular)

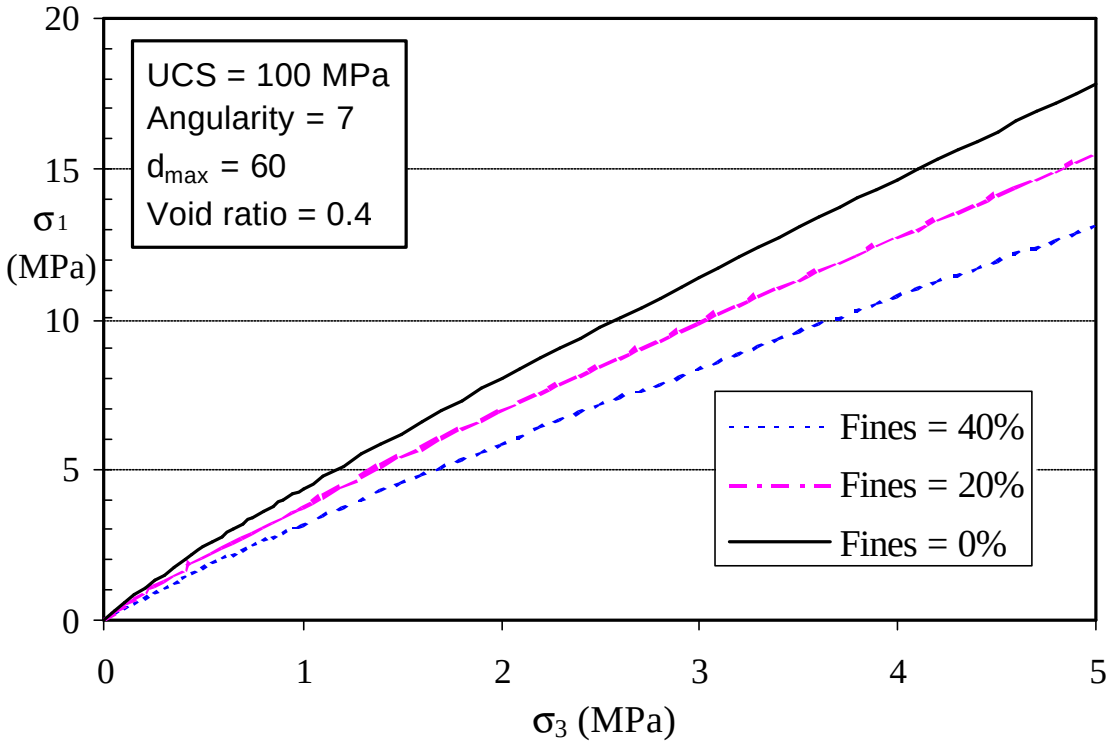


Figure 4.39. Effect of fines content on s¢1

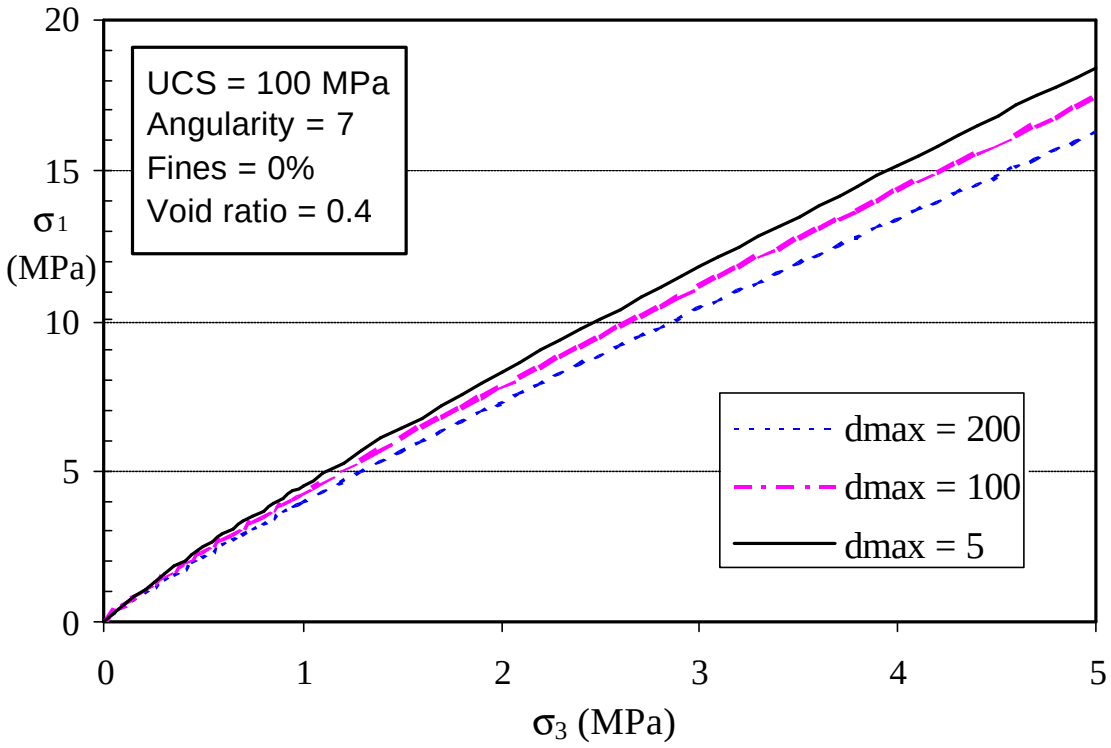


Figure 4.40. Effect of maximum particle size on s¢1

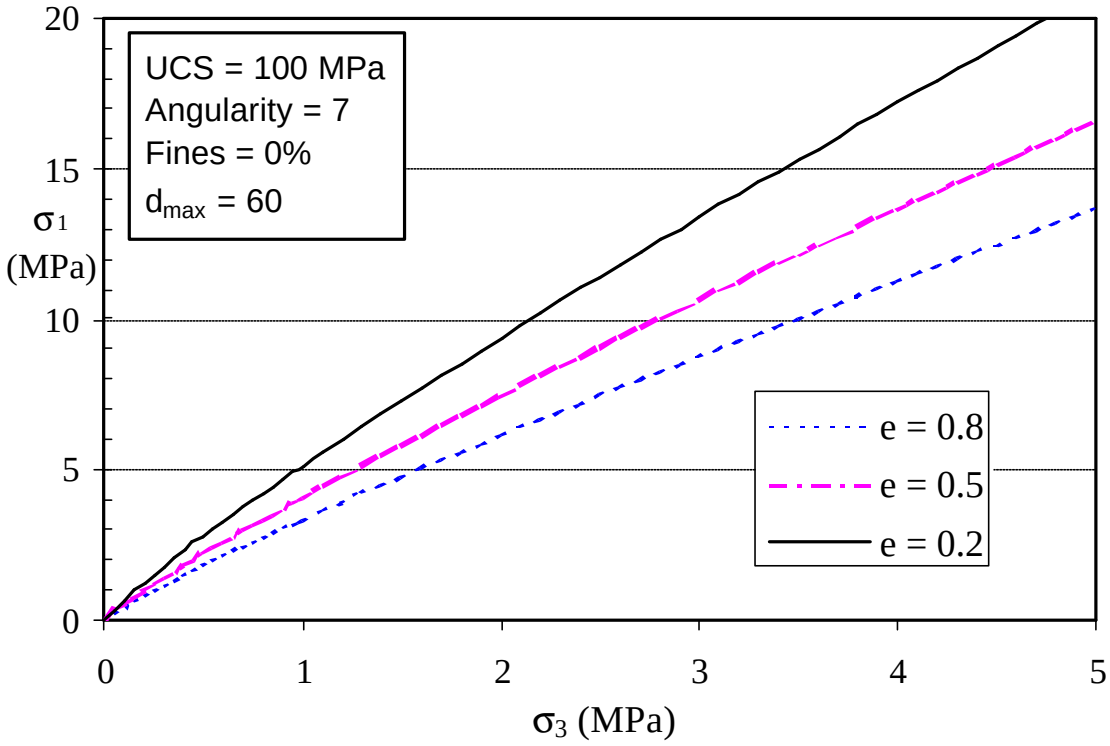


Figure 4.41. Effect of initial void ratio on s_1

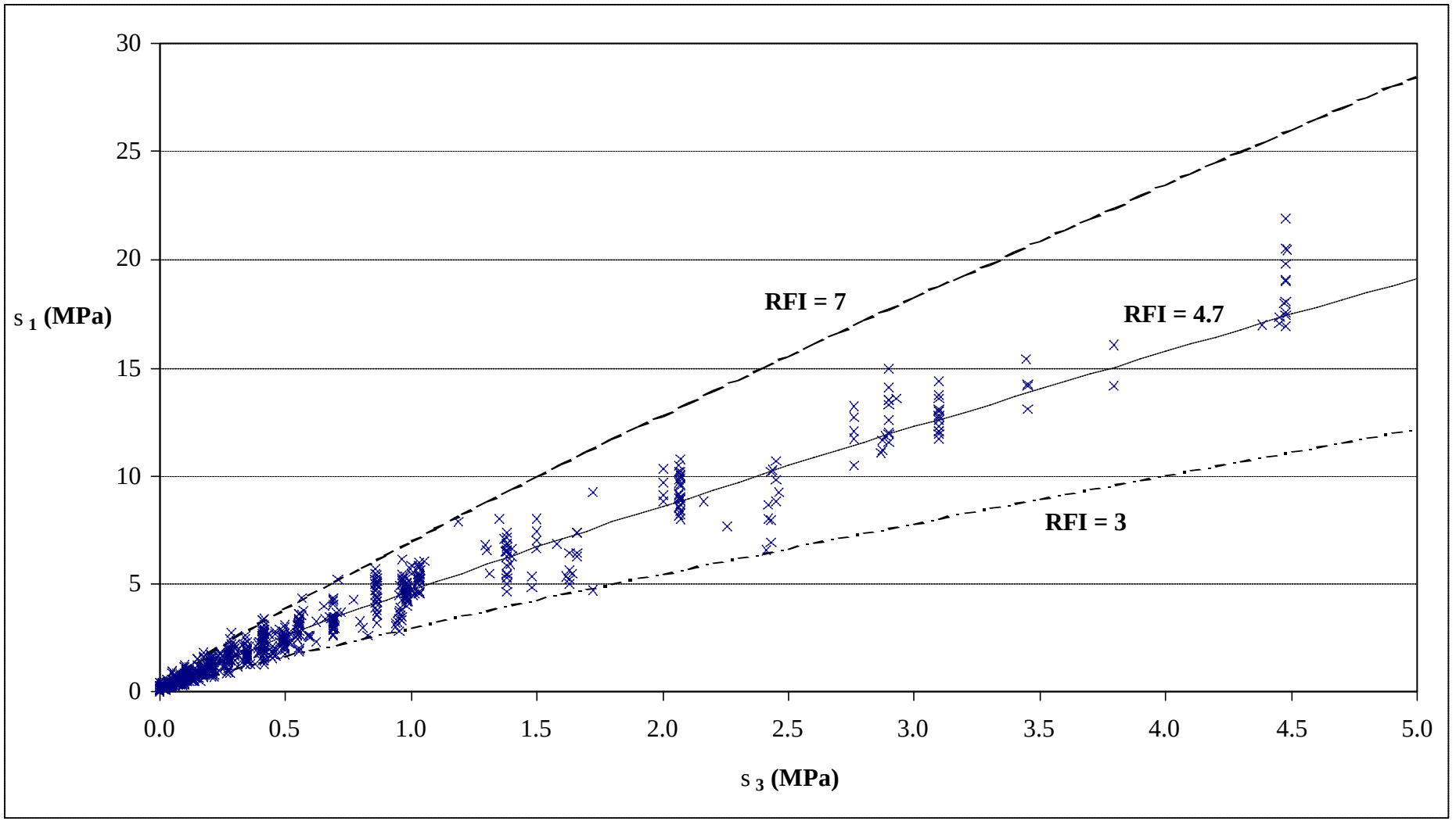


Figure 4.42. s'_1 **vs** s'_3 **showing data used in analysis and RFI relationship**

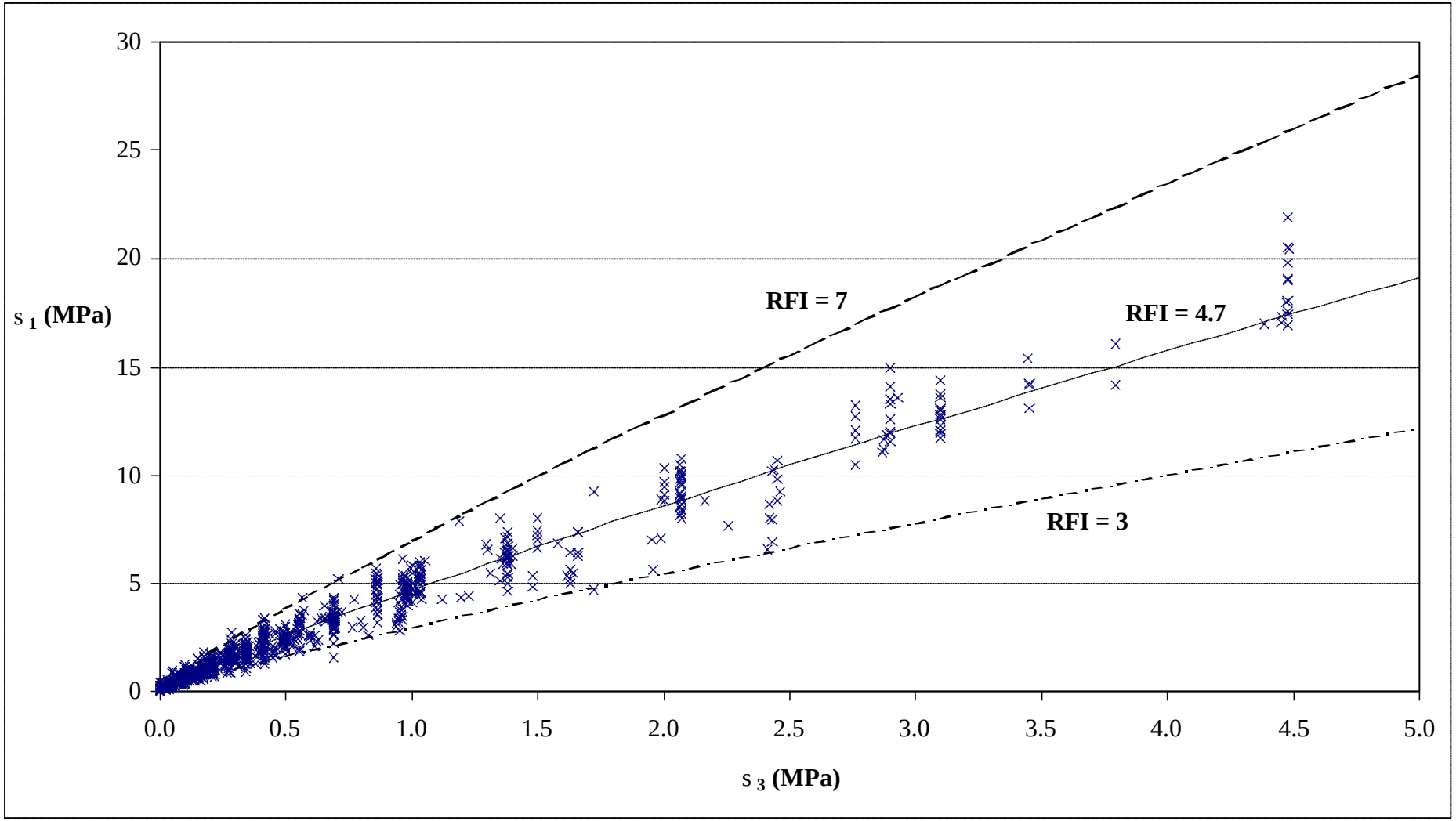


Figure 4.43. s'_1 **vs** s'_3 **showing all data and RFI relationship**

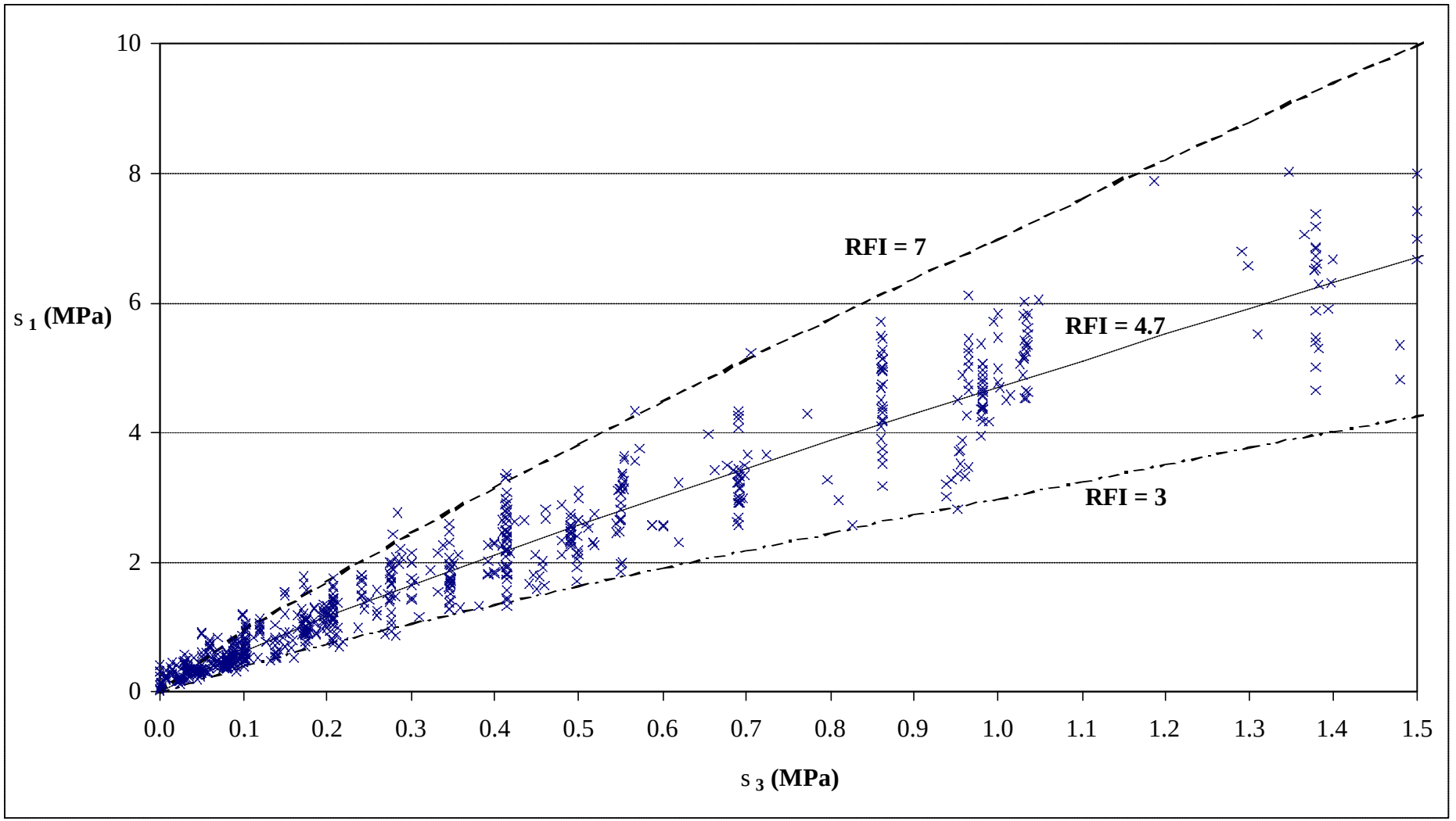


Figure 4.44. s'_1 **vs** s'_3 **showing all data and RFI relationship (**s'_3 **up to 1.5MPa)**

4.5.3.1 Secant Friction Angle Versus Normal Stress

Figure 4.37 to Figure 4.41 are replotted as curves for ϕ_{sec} vs σ_n by using the following equations:

$$f_{sec} = \sin^{-1}\left(\frac{RFI\, s_3'^{a} - s_3'}{RFI\, s_3'^{a} + s_3'}\right) \tag{4.31}$$

$$s_n = \cos f_{sec}\sqrt{RFI\, s_3'^{a+1}} \tag{4.32}$$

Table 4.8 shows the variation in ϕ_{sec} due to changes in various parameters for σ_n=1MPa and 0.5MPa. These give some idea of the influence of different parameters on the secant friction angle. Note that these friction angles are based on assumptions as shown on Figure 4.45 to Figure 4.49. The values chosen for the parameters are realistic upper and lower bounds from the database.

Table 4.8. Changes in f_{sec} on Figure 4.45 to Figure 4.49 for s_n=1MPa and s_n = 0.5MPa

Parameter	Value	σ_n = 0.1MPa		σ_n = 0.5MPa		σ_n = 1MPa	
		ϕ_{sec} (°)	Change in ϕ with increase in parameter	ϕ_{sec} (°)	Change in ϕ with increase in parameter	ϕ_{sec} (°)	Change in ϕ with increase in parameter
σ_c (MPa)	50	46.8	+1.3	42.2	+1.4	40.3	+1.5
	200	48.1		43.6		41.8	
Angularity rating	1-7	46.7	+2.1	42.2	+2.2	40.3	+2.3
	8	48.8		44.4		42.6	
Fines content (%)	0	46.7	-2.8	42.2	-3.1	40.3	-3.4
	20	43.9		39.1		36.9	
Maximum particle size (mm)	50	47.0	-2.2	42.4	-2.1	40.4	-2.1
	200	44.8		40.3		38.3	
Initial void ratio	0.2	49.9	-6.4	45.5	-9.1	43.7	-9.5
	0.6	43.5		36.4		34.2	

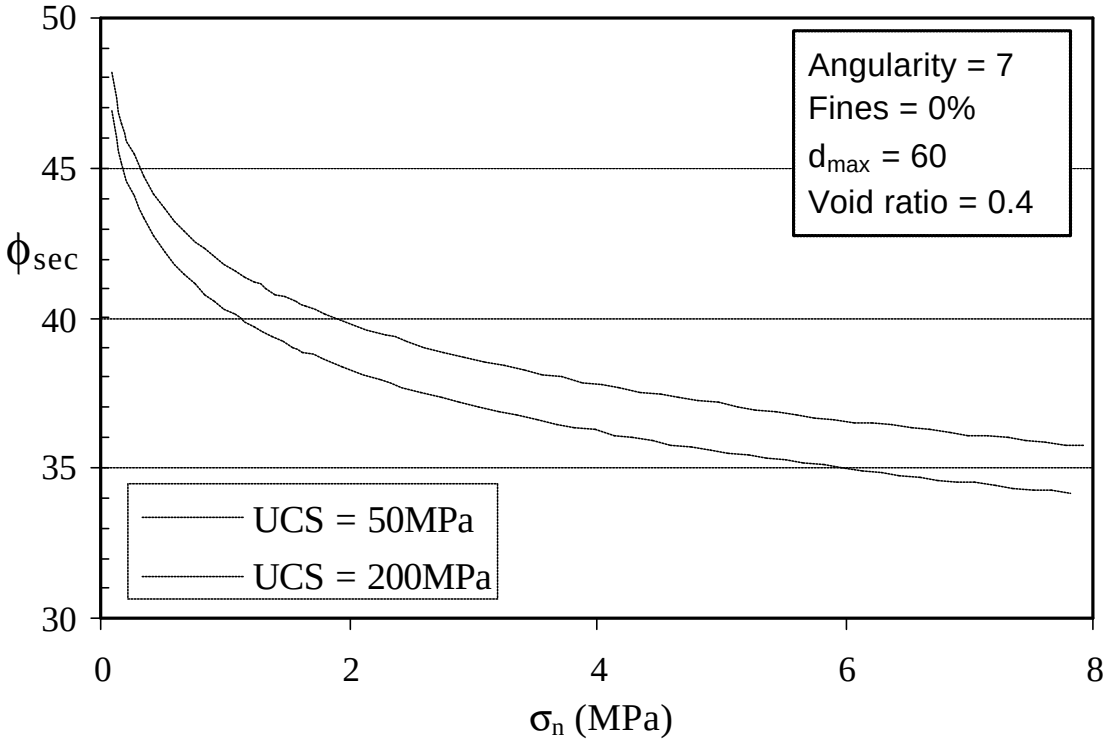


Figure 4.45. Effect of unconfined compressive strength on f_{sec}

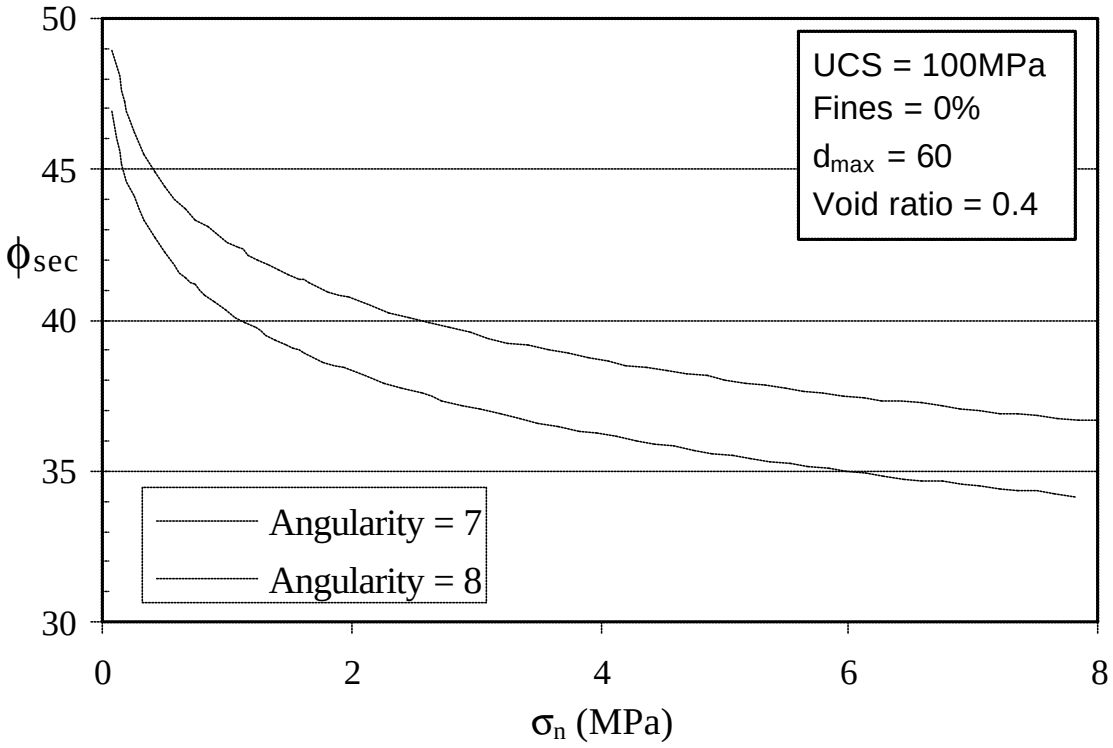


Figure 4.46. Effect of angularity on f_{sec} (7 = sub-angular to angular; 8 = angular)

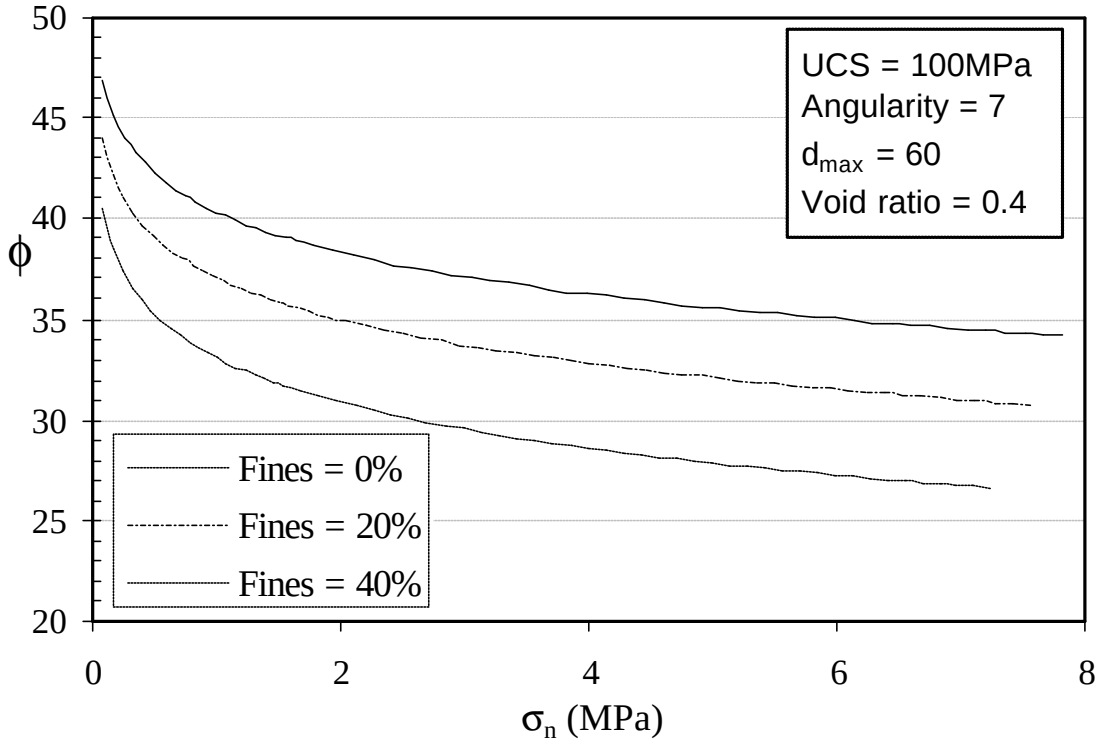


Figure 4.47. Effect of fines content on f_{sec}

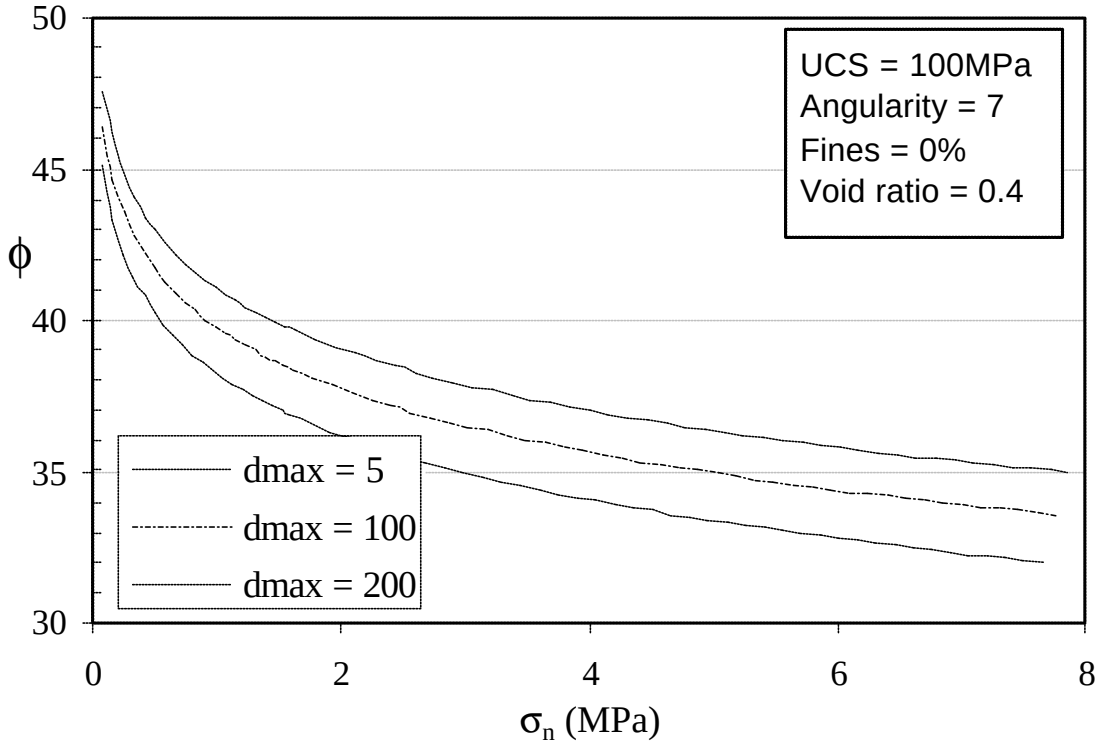


Figure 4.48. Effect of maximum particle size on f_{sec}

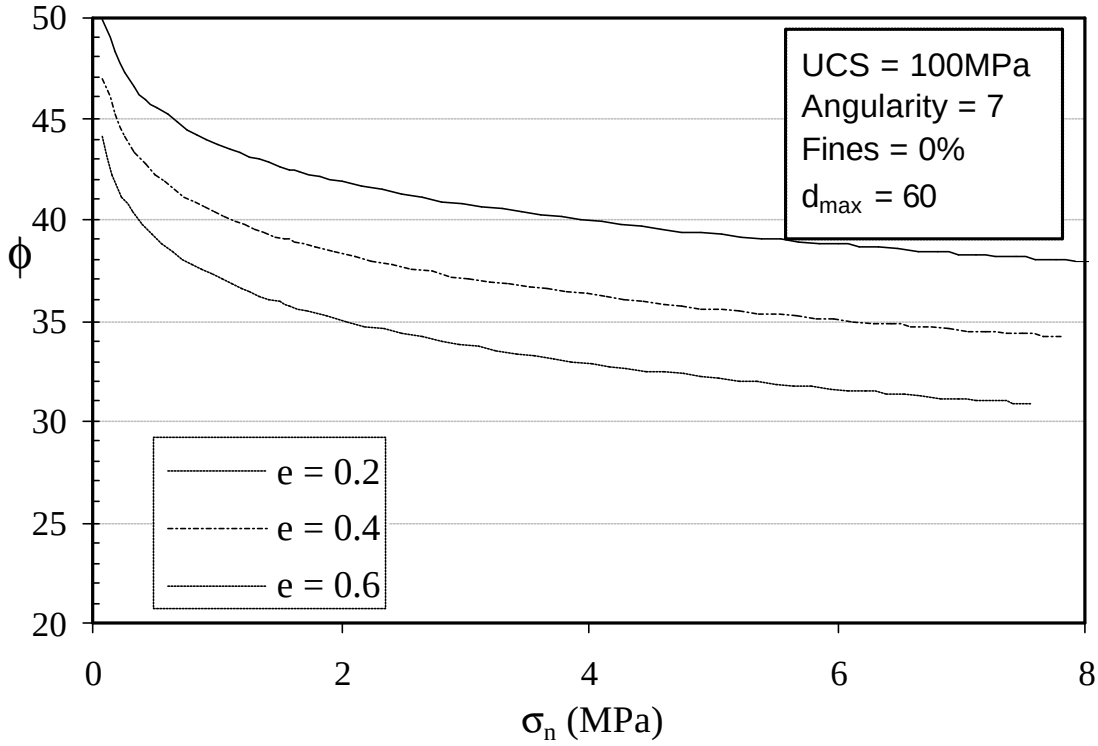


Figure 4.49. Effect of initial void ratio on f_{sec}

4.5.4 Hoek-Brown Criterion

The author considers that rockfill could represent a very poor quality rockmass. As such an analysis of the database using the Hoek-Brown criterion was performed. The results from this analysis could then be used as a lower bound rock mass strength estimator.

The statistical analysis was carried out for data where the UCS was known. This resulted in a sample of 409 triaxial tests. Two approaches were used: one where the variables are σ'_3 and the dependent variable σ'_1; and the other where the variables are σ'_3/σ_c and the dependent variable σ'_1/σ_c. Similar analyses to those discussed previously were carried out i.e. non-linear, quasi-Newton with a loss function of (observed – predicted)^2. A further analysis was carried out where the estimated values of the UCS of the intact rock were used where no UCS was recorded for the sample.

The ability of the Hoek-Brown criterion to estimate the shear strength of rockfill (and subsequently poor quality rock mass) was performed by assuming typical Hoek-Brown parameters of $s = 0$ and $a = 0.6$. This was compared with a further analyses where a was defined as a parameter. The results from the analyses are shown in Table 4.9.

Table 4.9 and Figure 4.50 show that if the recommended Hoek-Brown parameters are used a poor fit results. Due to the enforced curvature (a = 0.6), σ'_1 is overpredicted at low σ'_3 and under predicted at high σ'_3. Much better fits (Table 4.9 and Figure 4.50) are obtained where a was a parameter. These analyses showed that an a of 0.90 to 0.95 was much more suitable for rockfill. Similarly to intact rock, the interrelation between a and m_b is also illustrated by the results in Table 4.9.

The conclusion from the point of view of rock masses is that a is too constrained in the current Hoek-Brown criterion. Modifications need to be made such that the exponent, a, towards a value of 0.90-0.95 for poor quality rockfill.

A look at the forms of the equations used shows that the method of fitting is very important to the results obtained. In this case, non-dimensionalising the equations resulted in a better variance explained.

Table 4.9. Results from the statistical analysis of the rockfill database using the Hoek-Brown equation

Form of equation	Data sets	m_b	a	Variance explained
$s'_1 = s'_3 + s_c\left(\frac{m_b s'_3}{s_c} + 0\right)^{0.6}$	405 σ_{ci} known	0.42929	0.6 (input)	88.95%
$s'_1 = s'_3 + s_c\left(\frac{m_b s'_3}{s_c} + 0\right)^{a}$	405 σ_{ci} known	2.40706	0.89999	97.40%
$\frac{s'_1}{s_c} = \frac{s'_3}{s_c} + \left(m_b \frac{s'_3}{s_c} + 0\right)^{0.6}$	405 σ_{ci} known	0.75486	0.6 (input)	90.93%
$\frac{s'_1}{s_c} = \frac{s'_3}{s_c} + \left(m_b \frac{s'_3}{s_c} + 0\right)^{a}$	405 σ_{ci} known	2.71833	0.94877	98.73%
$s'_1 = s'_3 + s_c\left(\frac{m_b s'_3}{s_c} + 0\right)^{a}$	988 σ_{ci} estimated where unknown	2.39551	0.8954	97.42%

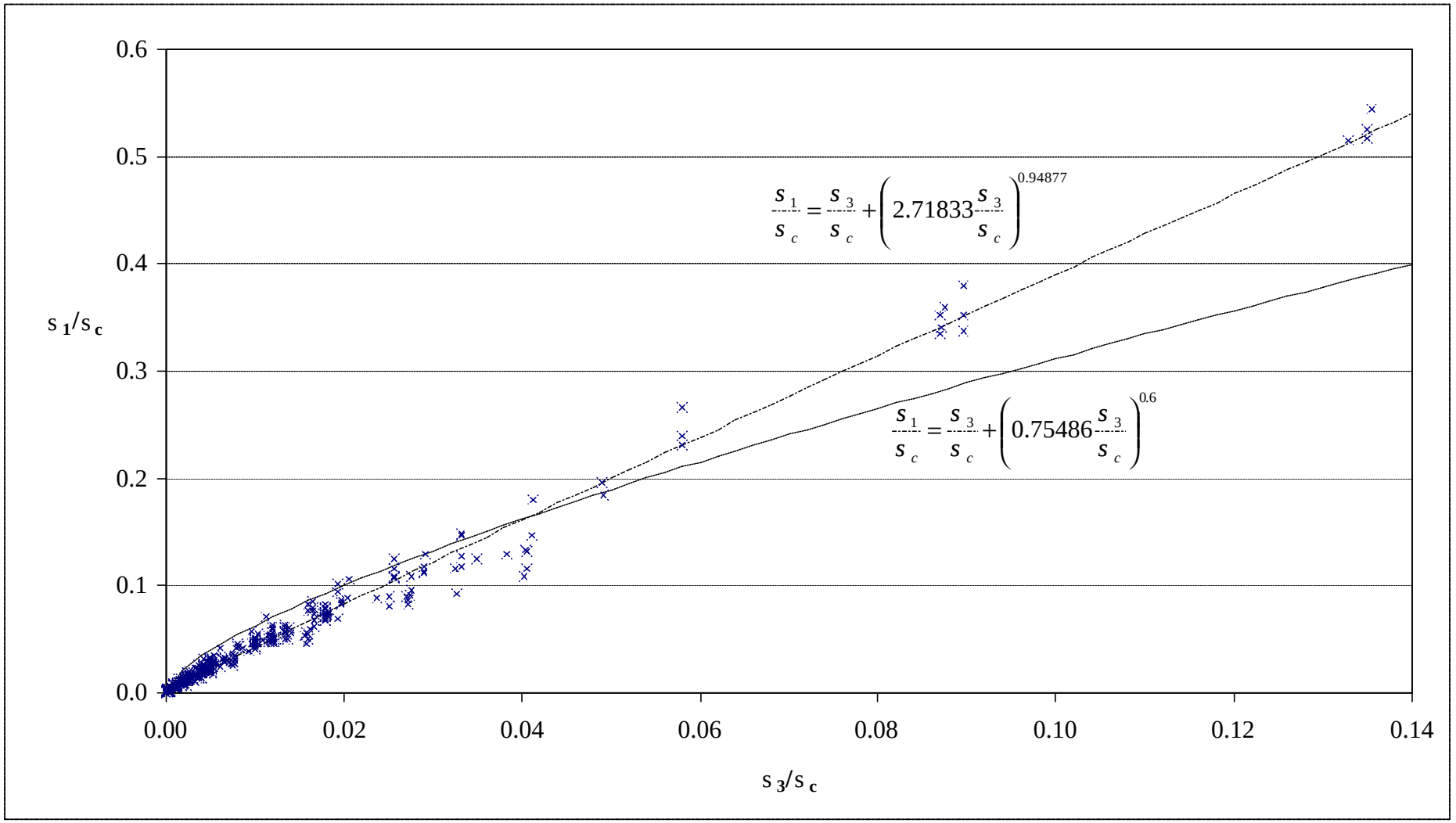


Figure 4.50. Statistical analysis results using Hoek-Brown formula and for *a* = 0.6 and *a* = 0.95

4.6 CONCLUSION

This chapter presented an overview of the strength of rockfill. An analysis of a large database of test results was used to develop two new shear strength equations, one relating the secant friction angle and normal stress and the other the principal stresses. The equation for principal stresses provided a much better fit to the data and is recommended. Of the parameters statistically investigated, the unconfined compressive strength, particle angularity, fines content, maximum particle size and void ratio were found to have the most significant effect on the shear strength of rockfill.

An analysis using the database and the Hoek-Brown criterion demonstrated that the criterion provided a poor fit to the rockfill data due to the restrictions placed on the exponent *a* of approximately 0.6. A good fit was obtained where *a* was free to vary up to unity (a value of 0.9 to 0.95 was obtained). This has important implications for the use of the Hoek-Brown criterion for very poor quality rock masses.